



OEM6 to OEM7 Integration Guide

OEM6 to OEM7 Integration Guide

Publication Number: D100401

Revision Level: v2

Revision Date: May 2025

Firmware Version: 7.09.05 / OM7MR0905RN0000

Proprietary notice

This document and the information contained herein are the exclusive properties of the legal entities that comprise the Autonomous Solutions division of the Hexagon AB group of companies ("Hexagon").

No part of this document may be reproduced, displayed, distributed, or used in any medium, in connection with any other materials, or for any purpose without prior written permission from Hexagon. Applications for permission may be directed to contact.ap@hexagon.com. Unauthorized reproduction, display, distribution or use may result in civil as well as criminal sanctions under the applicable laws. Hexagon aggressively protects and enforces its intellectual property rights to the fullest extent allowed by law.

This document and the information contained herein are provided AS IS and without any representation or warranty of any kind. Hexagon disclaims all warranties, express or implied, including but not limited to any warranties of merchantability, non-infringement, and fitness for a particular purpose. Nothing herein constitutes a binding obligation on Hexagon.

The information contained herein is subject to change without notice.

ALIGN, GLIDE, NovAtel, NovAtel CORRECT, OEM6, OEM7, PwrPak7, RTK ASSIST, SPAN, STEADYLINE and TerraStar are trademarks of Hexagon AB and/or its subsidiaries and affiliates, and/or their licensors. All other trademarks are properties of their respective owners.

© Copyright 2017 – 2025 Hexagon AB and/or its subsidiaries and affiliates. All rights reserved.

Table of Contents

Figures

Tables

Customer Support

Chapter 1 OEM6 to OEM7 integration

Chapter 2 New for OEM7 – firmware

2.1 Firmware version naming	14
2.2 OEM6 to OEM7 – commands	14
2.2.1 Deleted commands	14
2.2.2 Revised commands	16
2.2.3 New commands	18
2.3 OEM6 to OEM7 – logs	20
2.3.1 Deleted logs	20
2.3.2 Revised logs	21
2.3.3 New logs	23
2.4 OEM6 to OEM7 – SPAN	25
2.4.1 Translational offsets	25
2.4.2 Rotational offsets	25
2.4.3 INS profiles	26
2.4.4 INS seed	26
2.4.5 ALIGN calibration	26
2.4.6 Multi-line body to vehicle calibration	26
2.4.7 IMU-KVH1750 and IMU-P1750 baud rate conversion	27
2.4.8 INS Profiles	28
2.4.9 Multi-line body to vehicle frame rotation calibration routine	29
2.4.10 INS Seed / Fast INS initialization	31

Chapter 3 OEM6 to OEM7 – hardware

3.1 Physical changes	34
3.1.1 Shielding	34
3.1.2 Mounting	34
3.1.3 Weight	34
3.2 Electrical changes	35
3.2.1 COM port data rate	35
3.2.2 Power supply	35
3.2.3 USB interfaces	36
3.2.4 Input and output lines	36
3.2.5 LNA power	36
3.3 Environmental changes	37
3.3.1 Thermal dissipation	37
3.3.2 Vibration	37
3.4 Interference	37
3.4.1 Interference from co-located LNA	38

3.1 Receiver card interface examples	39
3.4.2 EVENT_IN, EVENT_OUT and PPS signal protection	39
3.4.3 Position Valid (PV) LED driver	41
3.4.4 Communication ports	42
3.4.5 CAN controller ports	43
3.4.6 USB interface	44
3.4.7 Ethernet port	46
3.5 Differences between PwrPak7 and OEM6 enclosures	48
3.6 Differences between SMART7 and SMART6	52
3.7 Differences between RELAY7 and RELAY	55

Chapter 4 Receiver technical specifications

Figures

Figure 1: Multi-line IMU body to vehicle calibration30

Figure 2: SMART antenna interference example 38

Figure 3: Protection and buffering for EVENT_IN, PPS and EVENT_OUT signals 40

Figure 4: OEM7 buffer for driving high-brightness LEDs from PV42

Figure 5: OEM7 CAN transceiver example 43

Figure 6: OEM7 USB device interface example44

Figure 7: OEM7 USB host interface example 45

Figure 8: Ethernet reference schematic47

Tables

Table 1: Deleted and replaced commands in OEM7	14
Table 2: Revised commands in OEM7	16
Table 3: New commands in OEM7	18
Table 4: Deleted and replaced logs in OEM7	20
Table 5: Revised logs in OEM7	21
Table 6: New logs in OEM7	23
Table 7: Commands for entering SPAN offsets and rotations	25
Table 8: OEM7 INS Profiles	28
Table 9: NVM seed indication	33
Table 10: OEM6 receiver inrush current	35
Table 11: OEM7 receiver inrush current	35
Table 12: EVENT_IN, EVENT_OUT and PPS pin designation	40
Table 13: Bill of materials (critical components)	41
Table 14: PV pin designation	42
Table 15: PV LED driver bill of materials (critical components)	42
Table 16: CAN transceiver pin designation	43
Table 17: CAN transceiver example bill of materials (critical components)	43
Table 18: USB device interface pin designation	44
Table 19: USB device interface example bill of materials	45
Table 20: USB host interface pin designation	46
Table 21: USB host interface example bill of materials	46
Table 22: Ethernet pin designation	47
Table 23: Ethernet transformer characteristics	47
Table 24: Bill of materials (critical components)	48
Table 25: Physical specifications	48
Table 26: Power requirements	48
Table 27: Signals tracked	48
Table 28: Horizontal position accuracy (RMS)	49
Table 29: Communications ports	50
Table 30: RF input	51
Table 31: I/O strobes	51
Table 32: Features	51
Table 33: Physical specifications	52
Table 34: Power requirements	52

Table 35: Signals tracked	52
Table 36: Horizontal position accuracy	53
Table 37: Communications ports	54
Table 38: I/O strobes	54
Table 39: Features	55
Table 40: Physical specifications	55
Table 41: Power requirements	55
Table 42: Supported radios	56

Customer Support

NovAtel knowledge base

If you have a technical issue, visit the NovAtel Support page at novatel.com/support. Through the *Support* page you can contact Customer Support, find papers and tutorials or download the latest firmware. To access the latest user documentation, visit docs.novatel.com/OEM7.

Before contacting Customer Support

Before contacting NovAtel Customer Support about a software problem, perform the following steps:



If logging data over an RS-232 serial cable, ensure that the configured baud rate can support the data bandwidth (see **SERIALCONFIG** command). NovAtel recommends a minimum suggested baud rate of 230400 bps.

1. Use the information in the **Troubleshooting** section to diagnose and troubleshoot your receiver's symptoms. Refer to the OEM7 Documentation Portal (docs.novatel.com/OEM7/Content/Troubleshooting/Troubleshooting.htm).
2. Log the data suggested in the appropriate *Troubleshooting Logs* section to a file on your computer for 15 minutes.
 - *General troubleshooting logs* on the next page
 - *Tracking and interference troubleshooting logs* on the next page
 - *SPAN troubleshooting logs* on page 10
 - *RTK troubleshooting logs* on page 10
 - *PPP troubleshooting logs* on page 11
 - *ALIGN troubleshooting logs* on page 12

If using NovAtel Application Suite, log the Troubleshooting message set for 15 minutes.

3. Send the data file to NovAtel Customer Support: support.novatel@hexagon.com
4. You can also issue a **FRESET** command to the receiver to clear any unknown settings.



The **FRESET** command will erase all user settings. You should know your configuration (by requesting the RXCONFIGA log) and be able to reconfigure the receiver before you send the **FRESET** command.

If you are having a hardware problem, send a list of the troubleshooting steps taken and the results.

Contact information

Log a support request with NovAtel Customer Support using one of the following methods:

Log a case and search knowledge:

Website: novatel.com/support

Log a case, search knowledge and view your case history: (login access required)

Web Portal: shop.novatel.com/novatelstore/s/login/

E-mail:

support.novatel@hexagon.com

Telephone:

U.S. and Canada: 1-800-NOVATEL (1-800-668-2835)

International: +1-403-295-4900

General troubleshooting logs

```
LOG RXSTATUSB ONCHANGED
LOG RAWEPHEMB ONNEW
LOG GLORAWEPHEMB ONNEW
LOG BESTPOSB ONTIME 1
LOG RANGE B ONTIME 0.5
LOG RXCONFIGA ONCE
LOG VERSIONA ONCE
LOG LOGLISTA ONCE
LOG PORTSTATSA ONTIME 10
LOG PROFILEINFOA ONCE
LOG HWMONITORA ONTIME 10
```

Tracking and interference troubleshooting logs

```
LOG VERSIONA ONCE
LOG RXCONFIGA ONCE
LOG CHANCONFIGLISTB ONCE
LOG PASSTHROUGH A ONNEW
LOG RXSTATUSB ONCHANGED
LOG CLOCKSTEERINGB ONCHANGED
LOG RAWEPHEMB ONNEW
LOG GLORAWEPHEMB ONNEW
LOG GALINAVRAWEPHEMERISB ONNEW
LOG BDSEPHEMERISB ONNEW
LOG QZSSEPHEMERISB ONNEW
LOG NAVICEPHEMERISB ONNEW
LOG RAWALMB ONNEW
LOG GLORAWALMB ONNEW
LOG GALALMANACB ONNEW
LOG BDSALMANACB ONNEW
LOG QZSSALMANACB ONNEW
LOG NAVICALMANACB ONNEW
LOG IONUTC B ONNEW
LOG GLOCLOCKB ONNEW
LOG GALCLOCKB ONNEW
LOG BDSCLOCKB ONNEW
LOG TRACKSTATB ONTIME 1
LOG RANGE B ONTIME 0.5
LOG BESTPOSB ONTIME 1
LOG SATVIS2B ONTIME 30
LOG ITDETECTSTATUSB ONCHANGED
```

For dual antenna receivers, add these logs:

```
LOG TRACKSTATB_1 ONTIME 1
LOG RANGE_1 ONTIME 0.5
```

For interference issues add this log:

```
LOG ITPSDDETECTB ONNEW
```

For interference issues, when you have enough datalink bandwidth to handle large logs, add this log:

```
LOG ITPSDFINALB ONNEW
```

SPAN troubleshooting logs

```
LOG RXSTATUSB ONCHANGED
LOG RAWEPHEMB ONNEW
LOG GLORAWEPHEMB ONNEW
LOG GALINAVRAWEPHEMERISB ONNEW
LOG BDSEPHemerisB ONNEW
LOG BESTPOSB ONTIME 1
LOG RANGE_1 ONTIME 0.5
LOG RXCONFIGA ONCE
LOG VERSIONA ONCE
LOG RAWIMUSXB ONNEW
LOG INSUPDATESTATUSB ONCHANGED
LOG INSPVAXB ONTIME 1
LOG INSCONFIGA ONCHANGED
LOG BESTGNSSPOSB ONTIME 1
```

RTK troubleshooting logs

```
LOG RXSTATUSB ONCHANGED
LOG RAWEPHEMB ONNEW
LOG GLORAWEPHEMB ONNEW
LOG QZSSRAWEPHEMB ONNEW
LOG BDSRAWNAVSUBFRAMEB ONNEW
LOG GALFNAVRAWEPHEMERISB ONNEW
LOG GALINAVRAWEPHEMERISB ONNEW
LOG RANGE_1 ONTIME 0.5
LOG BESTPOSB ONTIME 1.0
LOG RXCONFIGB ONCE
LOG VERSIONB ONCE
LOG TRACKSTATB ONTIME 1.0
LOG RTKPOSB ONTIME 1.0
LOG MATCHEDPOSB ONNEW
LOG MATCHEDSATSB ONNEW
LOG RTKSATSB ONTIME 1.0
LOG PSRPOSB ONTIME 1.0
LOG RAWALMB ONNEW
LOG IONUTCB ONNEW
LOG GLORAWALMB ONNEW
LOG GLOCLOCKB ONNEW
LOG PASSTHROUGHB ONNEW
LOG CLOCKMODEL ONTIME 1.0
LOG REFSTATIONB ONNEW
LOG RTKVELB ONTIME 1.0
```

PPP troubleshooting logs

```
LOG RXSTATUSB ONCHANGED
LOG GPSEPHEMB ONNEW
LOG GLOEPHEMERISB ONNEW
LOG QZSSEPHEMERISB ONNEW
LOG BDSEPHEMERISB ONNEW
LOG BDSBCNAV1EPHEMERISB ONNEW (firmware versions 7.08.03 and 7.08.10 and later)
LOG BDSBCNAV2EPHEMERISB ONNEW (firmware versions 7.08.03 and 7.08.10 and later)
LOG BDSBCNAV3EPHEMERISB ONNEW (firmware versions 7.08.03 and 7.08.10 and later)
LOG GALFNAVEPHEMERISB ONNEW
LOG GALINAVEPHEMERISB ONNEW
LOG RANGE B ONTIME 0.5
LOG BESTPOS B ONTIME 1.0
LOG RXCONFIG B ONCE
LOG VERSION B ONCE
LOG TRACKSTAT B ONTIME 10.0
LOG LBANDTRACKSTAT B ONTIME 1.0
LOG PPPPOS B ONTIME 1.0
LOG PPPSAT B ONTIME 1.0
LOG TERRASTARINFO B ONCHANGED
LOG TERRASTARSTATUS B ONCHANGED
LOG PSRPOS B ONTIME 1.0
LOG ALMANAC B ONNEW
LOG GLOALMANAC B ONNEW
LOG GALALMANAC B ONNEW
LOG BDSALMANAC B ONNEW
LOG QZSSALMANAC B ONNEW
LOG IONUTC B ONNEW
LOG GLOCLOCK B ONNEW
LOG LBANDBEAMTABLE B ONCHANGED
```

ALIGN troubleshooting logs

```
LOG RXSTATUSB ONCHANGED
LOG RAWEPHEMB ONNEW
LOG GLORAWEPHEMB ONNEW
LOG GALINAVRAWEPHEMERISB ONNEW
LOG BDSEPHemerISB ONNEW
LOG BESTPOSB ONTIME 1
LOG RANGE B ONTIME 0.5
LOG RXCONFIGA ONCE
LOG VERSIONA ONCE
LOG LOGLISTA ONCE
LOG PORTSTATSA ONTIME 10
LOG ALIGNBSLNENUB ONNEW
LOG ALIGNBSLNXYZB ONNEW
LOG ALIGNDOPB ONNEW
LOG HEADING2B ONNEW
LOG MASTERPOSB ONNEW
LOG ROVERPOSB ONNEW (This log can only be output on a receiver where ALIGN is enabled.)
LOG HEADINGSATSA ONNEW (This log can only be output on a receiver where ALIGN is enabled and
configured as the ALIGN rover.)
LOG DUALANTENNAHEADINGB ONTIME 1.0
LOG TRACKSTATB_1 ONTIME 1 (dual antenna receivers only)
LOG RANGE B_1 ONTIME 0.5 (dual antenna receivers only)
```

Chapter 1 OEM6 to OEM7 integration

When upgrading from an OEM6 receiver to an OEM7 receiver, there are several hardware and software differences that must be accounted for in the system design.

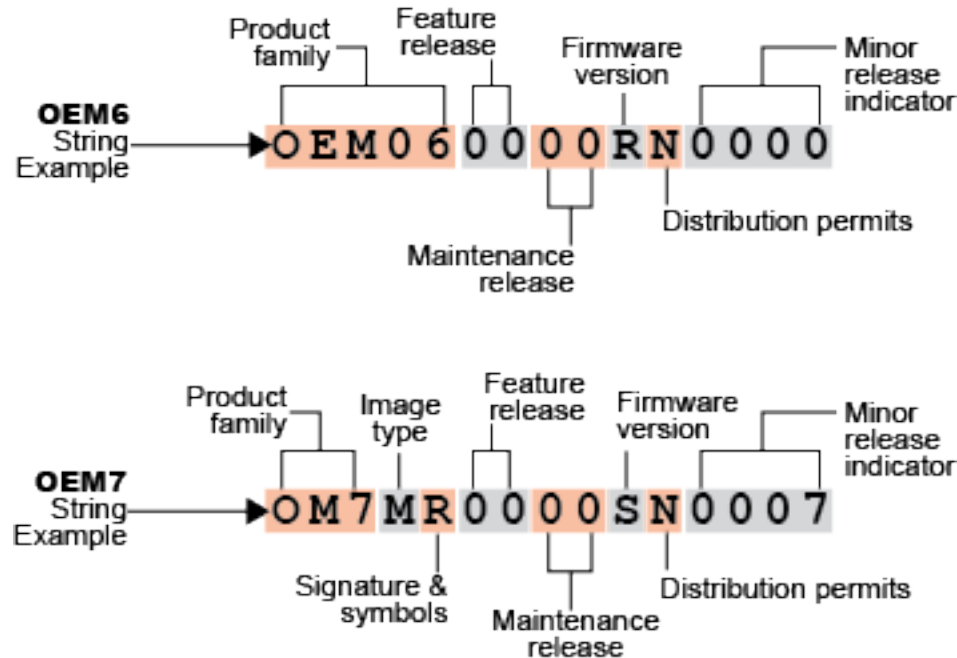
These differences are described in the following chapters:

- *New for OEM7 – firmware* on page 14
- *OEM6 to OEM7 – hardware* on page 34
- *Receiver technical specifications* on page 57

Chapter 2 New for OEM7 – firmware

This chapter describes the new commands, logs, features and functionality available on OEM7 receivers.

2.1 Firmware version naming



For information about the fields in the firmware version, refer to the **VERSION** log.

2.2 OEM6 to OEM7 – commands

The following sections describe the command changes between OEM6 and OEM7.

For information about the OEM7 commands, refer to the [OEM7 Commands and Logs Reference Manual](#).

2.2.1 Deleted commands

The following table lists the OEM6 commands that have been deleted or replaced in OEM7.

Table 1: Deleted and replaced commands in OEM7

OEM6	OEM7
APPLYVEHICLEBODYROTATION	Deleted.
ASSIGNLBAND	Replaced. Use the existing ASSIGNLBANDBEAM command
ASSIGNLBAND2	Replaced. Use the existing ASSIGNLBANDBEAM command
BASEANTENNAMODEL	Replaced. Use the existing BASEANTENNAPCO command and BASEANTENNAPCV command

OEM6	OEM7
COM	Replaced. Use the existing SERIALCONFIG command
COMCONFIG	Replaced. Use the existing SERIALCONFIG command, INTERFACEMODE command and ECHO command
DGPSEPHEMDELAY	Deleted.
EXTHDGOFFSET	Replaced. Use the new SETINSROTATION command
GIMBALSPANROTATION	Replaced. Use the new SETINSROTATION command
INSWHEELUPDATE	Replaced. Use the existing SETINSUPDATE command
INSZUPTCONTROL	Replaced. Use the existing SETINSUPDATE command
LEVERARMCALIBRATE	Replaced. Use the new INSCALIBRATE command
OMNIUSEGLONASS	Deleted.
PDPVELOCITYOUT	Replaced. Use the existing BESTVELTYPE command
RTKCOMMAND	Deleted.
RTKELEV MASK	Deleted.
RVBCALIBRATE	Replaced. Use the new INSCALIBRATE command
SETCANNNAME	Deleted.
SETGIMBALORIENTATION	Replaced. Use the new SETINSROTATION command
SETIMUORIENTATION	Replaced. Use the new SETINSROTATION command
SETIMUTOANTOFFSET	Replaced. Use the new SETINSTRANSLATION command
SETIMUTOANTOFFSET2	Replaced. Use the new SETINSTRANSLATION command
SETIMUTOEXTOFFSET	Replaced. Use the new SETINSTRANSLATION command
SETIMUTOGIMBALOFFSET	Replaced. Use the new SETINSTRANSLATION command
SETINSOFFSET	Replaced. Use the new SETINSTRANSLATION command
SETINITATTITUDE	Replaced. Use the existing SETINITAZIMUTH command
SETINSOFFSETS	Replaced. Use the new SETINSROTATION command and SETINSTRANSLATION command
SETINSROTATION	Replaced. The existing SETINSROTATION command (Message ID 1796) has removed. A new SETINSROTATION command (Message ID 1921) has been added. This new command has enhanced capability and is the recommend command to use.

OEM6	OEM7
SETMARK1OFFSET SETMARK2OFFSET SETMARK3OFFSET SETMARK4OFFSET	Replaced. Use the new SETINSROTATION command and SETINSTRLANSATION command
SETRTCM16	Deleted.
SETRTCM36	Deleted.
SETRTCMRXVERSION	Deleted.
SETRTCMTXVERSION	Deleted.
SETWHEELSOURCE	Deleted.
VEHICLEBODYROTATION	Replaced. Use the new SETINSROTATION command

2.2.2 Revised commands

Revised commands are listed in the following table.

Table 2: Revised commands in OEM7

OEM6	OEM7
ANTENNAPOWER	3V3 option removed  If a short circuit or other problem causes an overload of the current supplied to the antenna, the receiver hardware shuts down the power supplied to the antenna. To restore power, power cycle the receiver. The Receiver Status word, available in the RXSTATUS log, provides more information about the cause of the problem.
ASSIGN, ASSIGNALL, LOCKOUT, SBASCONTROL, TRACKSV, UNLOCKOUT	The OEM7 tracks SBAS PRNs 120-158 and 183-187. The OEM6 tracked 120-138 and 183-187
ASSIGNLBANDBEAM	This command now supports multiple L-Band channels. The Frequency field is entered only in Hz.

OEM6	OEM7		
AUTH	New options were added for the State parameter		
	ASCII	Binary	Description
	ERASE_ TABLE	7	Erase all auth codes from the system. Requires a special auth code to prevent against accidental erasing
	CLEAN_ TABLE	8	Remove all invalid auth codes from the system
	Special auth code for the ERASE_ TABLE case: PW5W2B,WW5TM9,WW2PCZ,WW3M4H,WW4HPG,ERASE_ AUTH Special auth code for the CLEAN_ TABLE case: 4DR69H,G369W8,34MNJJ,5NHXCJ,GW7C75,CLEAN_ AUTH		
COMCONTROL	The COM ports available have changed to reflect the OEM7 receivers.		
CONNECTIMU	Added the EPSON G320N IMU. Removed the Litef LCI-1 IMU. For the IMU Type parameter, the ASCII values for IMU no longer use the "IMU_" prefix. However, the legacy ASCII values that contain the "IMU_" prefix are still supported. The binary IMU type values have not changed.		
ETHCONFIG	Ethernet interface hardware now automatically connects properly to the other physical device		
EXTERNALPVAS	The format of several of the options in this command have been changed from Double to Float.		
FRESET	Added new FRESET targets.		
	Binary	ASCII	Description
	10	USERDATA	Resets the user data saved using the NVMUSERDATA command
INTERFACEMODE	Added new Interface Mode options:		
	Binary	ASCII	Description
	49	NOVATELMINBINARY	NovAtel binary message with a minimal header. Only available for CCOM ports.

OEM6	OEM7
LOG	The optional parameters Period, Offset and Hold are only used when the Trigger is ONTIME. Previous versions of the firmware accepted (but ignored) non-zero values for Period and Offset even if the Trigger was not ONTIME. The firmware on OEM7 receivers will reject such commands and return an error. It is especially important that these values are zero when entering a LOG command in Binary or ASCII format.
POSAVE	The “state” field no longer optional; default ON. A value must now be entered
SETINITAZIMUTH	The range for the standard deviation parameter has changed to 1 to 25 degrees.

2.2.3 New commands

The following table lists the commands added in OEM7.

Table 3: New commands in OEM7

OEM7	Description
CANCONFIG	Configures the CAN port parameters
CCOMCONFIG	Binds a CAN port to a J1939 node configures the CAN protocol used by the port.
FORCEGALE6CODE	Forces Galileo E6 channels to track E6B or E6C
INSCALIBRATE	Initiates the calibration of INS offsets
INSSEED	Enables or disables saving and restoring the last known SPAN solution
ITBANDPASSCONFIG	Configures a bandpass filter on the receiver
ITDETECTCONFIG	Enables interference detection
ITFRONTENDMODE	Configures the front end mode for each RF path
ITPROGFILTCONFIG	Configures filtering on the receiver
ITSPECTRALANALYSIS	Enables and configures spectral analysis on the receiver
J1939CONFIG	Configures the CAN J1939 network-level parameters
NMEAFORMAT	Customizes the NMEA output
NVMUSERDATA	Writes the data provided to NVM
PPPBASICCONVERGEDCRITERIA	Sets the convergence threshold for lower accuracy PPP solutions.
PPPDYNAMICSEED	Seed the PPP filter in any platform motion state
PPPRESET	Resets the PPP filter.
REFERENCESTATIONTIMEOUT	Sets a timeout for removing previously stored base stations
RFINPUTGAIN	Selects the mode of setting the CAG for the purpose of interference detection.

OEM7	Description
RTKPORTMODE	Assigns a port on the rover receiver for the RTK and ALIGN information being received
RTKRESET	Resets the RTK filter.
SETINSPROFILE	Set the filter behavior to optimize the output for a specific environment
SETINSROTATION	Specifies the rotational offsets between the IMU frame and other reference frames, such as the vehicle frame or an ALIGN baseline
SETINSTRANSFORMATION	Specifies the translational offsets between the IMU frame and other reference frames, including GNSS antennas or the desired output frame

2.3 OEM6 to OEM7 – logs

The following sections describe the log changes between OEM6 and OEM7.

For information about the OEM7 logs, refer to the [OEM7 Commands and Logs Reference Manual](#).

2.3.1 Deleted logs

The following table lists the OEM6 logs that have been deleted or replaced in OEM7.

Table 4: Deleted and replaced logs in OEM7

OEM6	OEM7
BESTLEVERARM	Replaced. Use the new INSCONFIG log
BESTLEVERARM2	Replaced. Use the new INSCONFIG log
CMR Data logs	Deleted the logs <i>CMRDESC</i> , <i>CMRGLOBS</i> , <i>CMROBS</i> , <i>CMRPLUS</i> , <i>CMRREF</i>
GALEPHEMERIS	Replaced with the GALFNAVEPHEMERIS log for Galileo FNAV ephemeris information and GALINAVEPHEMERIS log for Galileo INAV ephemeris information
GPGGARTK	Deleted Refer to the NMEAFORMAT command for information about accessing the information that was provided by this log.
HEADING	Replaced. Use the existing HEADING2 log
IMUTOANTOFFSETS	Replaced. Use the new INSCONFIG log
INSCOV	Replaced. Use the new INSSTDEV log
INSCOV5	Replaced. Use the new INSSTDEVS log
INSUPDATE	Replaced. Use the new INSUPDATESTATUS log
LBANDINFO	Replaced. Use the existing TERRASTARINFO log or VERIPOSINFO log to log subscription information Use the existing TERRASTARSTATUS log or VERIPOSSTATUS log to log decoder and subscription status information
LBANDSTAT	
OMNIHPPOS	
OMNIHPSATS	
OMNIVIS	
PSRTIME	Deleted
RAWLBANDFRAME	Deleted
RAWLBANDPACKET	Deleted
SATVIS	Replaced. Use the existing SATVIS2 log
RTCA Data Logs	Deleted the logs: <i>RTCA1</i> , <i>RTCAEPHEM</i> , <i>RTCAOBS</i> , <i>RTCAOBS2</i> , <i>RTCAOBS3</i> , <i>RTCAREF</i> , <i>RTCAREFEXT</i>

OEM6	OEM7
RTCM Data Logs	Deleted the logs: <i>RTCM1</i> , <i>RTCM3</i> , <i>RTCM9</i> , <i>RTCM15</i> , <i>RTCM16</i> , <i>RTCM16T</i> , <i>RTCM1819</i> , <i>RTCM2021</i> , <i>RTCM22</i> , <i>RTCM23</i> , <i>RTCM24</i> , <i>RTCM31</i> , <i>RTCM32</i> , <i>RTCM36</i> , <i>RTCM36T</i> , <i>RTCM59</i> , <i>RTCM59GLO</i> , <i>RTCMOMNI1</i>
WAAS0, WAAS1, WAAS2, etc.	Replaced. Use the existing SBAS0 log, SBAS1 log, SBAS2 log, etc.

2.3.2 Revised logs

Revised logs are listed in the table below.

Table 5: Revised logs in OEM7

OEM6	OEM7																																							
AUTHCODES	Added auth code types for Reserved and High Speed Signatures: 4 = RESERVED 5 = HIGH_SPEED																																							
CHANCONFIGLIST	The following signal types have been added to Table: CHANCONFIGLIST Signal Type																																							
	<table><tr><th>Value</th><th>Name</th><th>Description</th></tr><tr><td>32</td><td>BEIDOU B1B3</td><td>BeiDou B1, B3</td></tr><tr><td>33</td><td>BEIDOU B3</td><td>BeiDou B3</td></tr><tr><td>34</td><td>BEIDOU B1B2B3</td><td>BeiDou B1, B2 and B3</td></tr><tr><td>35</td><td>GALE1E5AE5BALTBOCE6</td><td>Galileo E1, E5A, E5B, AltBOC, E6</td></tr><tr><td>36</td><td>GPS L1L2PL2CL5L1C</td><td>GPS L1CA, L2P, L2C, L5, L1C</td></tr><tr><td>37</td><td>QZSS L1CAL2CL5L1C</td><td>QZSS L1CA, L2C, L5, L1C</td></tr><tr><td>38</td><td>QZSS L1CAL2CL5L1CL6</td><td>QZSS L1CA, L2C, L5, L1C, L6</td></tr><tr><td>39</td><td>GLONASS L1L3</td><td>GLONASS L1CA, L3</td></tr><tr><td>40</td><td>GLONASS L3</td><td>GLONASS L3</td></tr><tr><td>41</td><td>GLONASS L1L2PL2CL3</td><td>GLONASS L1CA, L2P, L2CA, L3</td></tr><tr><td>42</td><td>GPS L1L2PL2CL1C</td><td>GPS L1CA, L2P, L2C, L1C</td></tr><tr><td>43</td><td>QZSS L1CAL2CL1C</td><td>QZSS L1CA, L2C, L1C</td></tr></table>	Value	Name	Description	32	BEIDOU B1B3	BeiDou B1, B3	33	BEIDOU B3	BeiDou B3	34	BEIDOU B1B2B3	BeiDou B1, B2 and B3	35	GALE1E5AE5BALTBOCE6	Galileo E1, E5A, E5B, AltBOC, E6	36	GPS L1L2PL2CL5L1C	GPS L1CA, L2P, L2C, L5, L1C	37	QZSS L1CAL2CL5L1C	QZSS L1CA, L2C, L5, L1C	38	QZSS L1CAL2CL5L1CL6	QZSS L1CA, L2C, L5, L1C, L6	39	GLONASS L1L3	GLONASS L1CA, L3	40	GLONASS L3	GLONASS L3	41	GLONASS L1L2PL2CL3	GLONASS L1CA, L2P, L2CA, L3	42	GPS L1L2PL2CL1C	GPS L1CA, L2P, L2C, L1C	43	QZSS L1CAL2CL1C	QZSS L1CA, L2C, L1C
	Value	Name	Description																																					
	32	BEIDOU B1B3	BeiDou B1, B3																																					
	33	BEIDOU B3	BeiDou B3																																					
	34	BEIDOU B1B2B3	BeiDou B1, B2 and B3																																					
	35	GALE1E5AE5BALTBOCE6	Galileo E1, E5A, E5B, AltBOC, E6																																					
	36	GPS L1L2PL2CL5L1C	GPS L1CA, L2P, L2C, L5, L1C																																					
	37	QZSS L1CAL2CL5L1C	QZSS L1CA, L2C, L5, L1C																																					
	38	QZSS L1CAL2CL5L1CL6	QZSS L1CA, L2C, L5, L1C, L6																																					
	39	GLONASS L1L3	GLONASS L1CA, L3																																					
	40	GLONASS L3	GLONASS L3																																					
	41	GLONASS L1L2PL2CL3	GLONASS L1CA, L2P, L2CA, L3																																					
	42	GPS L1L2PL2CL1C	GPS L1CA, L2P, L2C, L1C																																					
43	QZSS L1CAL2CL1C	QZSS L1CA, L2C, L1C																																						
GPHDT	The GPHDT log can only be logged using the ONCHANGED trigger. Other triggers, such as ONTIME are not accepted.																																							
HWMONITOR	A new status, Peripheral Core Voltage, was added for OEM7 receivers. Also, the status options supported by OEM7 receivers are indicated.																																							

OEM6	OEM7																								
INSATTX, INSPOSX, INSPVAX, INSUPDATESTATUS, INSVELX, RELINSPVA, SYNCRELINSPVA	Additional status information has been added to the Extended Solution Status word. The Extended Solution Status word is available in these logs. See the INSATTX log for a description of the new status values available.																								
GIMBALLEDPVA, IMURATEPVA, IMURATEPVAS, INSATT, INSATTS, INSATTX, INSPOS, INSPOSS, INSPOSX, INSPVA, INSPVAS, INSPVAX, INSSPD, INSSPDS, INSVEL, INSVELS, INSVELX, MARKxPVA, RELINSPVA, SYNCRELINSPVA	The following statuses have been added to the Inertial Solution Status table. <table><tr><th>Binary</th><th>ASCII</th><th>Description</th></tr><tr><td>10</td><td>WAITING_ AZIMUTH</td><td>The INS filer has orientation, initial biases, initial position and valid roll/pitch estimated. Will not proceed until initial azimuth is entered.</td></tr><tr><td>11</td><td>INITIALIZING_ BIASES</td><td>The INS filter is estimating initial biases during the first 10 seconds of stationary data.</td></tr><tr><td>12</td><td>MOTION_ DETECT</td><td>The INS filter has not completely aligned, but has detected motion.</td></tr></table>					Binary	ASCII	Description	10	WAITING_ AZIMUTH	The INS filer has orientation, initial biases, initial position and valid roll/pitch estimated. Will not proceed until initial azimuth is entered.	11	INITIALIZING_ BIASES	The INS filter is estimating initial biases during the first 10 seconds of stationary data.	12	MOTION_ DETECT	The INS filter has not completely aligned, but has detected motion.								
Binary	ASCII	Description																							
10	WAITING_ AZIMUTH	The INS filer has orientation, initial biases, initial position and valid roll/pitch estimated. Will not proceed until initial azimuth is entered.																							
11	INITIALIZING_ BIASES	The INS filter is estimating initial biases during the first 10 seconds of stationary data.																							
12	MOTION_ DETECT	The INS filter has not completely aligned, but has detected motion.																							
LBANDTRACKSTAT	This log now reports the L-Band tracking and Viterbi status for more than one beam.																								
RANGECMP	GLONASS frequency number added to the Range Record Format table. The changes are shown below: <table><tr><th>Data</th><th>Bits first to last</th><th>Length (bits)</th><th>Scale factor</th><th>Units</th></tr><tr><td>C/No</td><td>165-169</td><td>5</td><td>(20+n)</td><td>dB-Hz</td></tr><tr><td>GLONASS Frequency Number</td><td>170-175</td><td>n+7</td><td>1</td><td></td></tr><tr><td>Reserved</td><td>176-191</td><td>16</td><td></td><td></td></tr></table>					Data	Bits first to last	Length (bits)	Scale factor	Units	C/No	165-169	5	(20+n)	dB-Hz	GLONASS Frequency Number	170-175	n+7	1		Reserved	176-191	16		
Data	Bits first to last	Length (bits)	Scale factor	Units																					
C/No	165-169	5	(20+n)	dB-Hz																					
GLONASS Frequency Number	170-175	n+7	1																						
Reserved	176-191	16																							
RANGE, RANGECMP, RANGECMP2	OEM7 tracks additional GNSS signals: GPS L1C, GLONASS L3, QZSS L1C and QZSS L6																								
RANGE, RANGECMP, RANGECMP2, RANGECMP4, RANGEGPSL1, SATVIS2, TRACKSTAT	The OEM7 tracks SBAS PRNs 120-158 and 183-192. The OEM7 tracks QZSS PRNs 193-202. The OEM6 tracks SBAS PRNs 120-138 and 183-187. The OEM6 tracks QZSS PRNs 193-197.																								

OEM6	OEM7
RXSTATUS	New OEM7 error and status states have been added to the Receiver Error, Receiver Status, Auxiliary 1 Status and Auxiliary 2 Status words. See the RXSTATUS log  The receiver family from which the RXSTATUS log was generated must be known to properly interpret the information provided by the Error and Status words. To determine the receiver family, use the Version Bits (25 and 26) of the Receiver Status word. If bit 25 is 1 and bit 26 is 0, the RXSTATUS log is from an OEM7 receiver. If bit 25 is 0 and bit 26 is 0, the RXSTATUS log is from an OEM6 or earlier receiver.
TRACKSTAT	The OEM7 has channel state 23 for BOC sidepeak check.
VARIABLELEVERARM	The computed lever arm offset is now reported in the IMU Body frame, rather than the SPAN computation frame.

2.3.3 New logs

The table below lists the logs added in OEM7.

Table 6: New logs in OEM7

OEM7	Description
GALCNAVRAWPAGE	Provides Galileo raw C-NAV page data from Galileo E6 signals.
INSCALSTATUS	Provides the status and estimated values for the currently running offset calibration.
INSCONFIG	Provides the current configuration of the SPAN system.
INSSTDEV	Provides the INS PVA standard deviations.
INSSTDEVS	Provides the INS PVA standard deviations and has a short header.
INSUPDATESTATUS	Provides the most recent INS update information
ITBANDPASSBANK	Provides the allowable bandpass filter configurations
ITDETECTSTATUS	Provides a list of detected interference
ITFILTTABLE	Provides the filter configuration for each frequency
ITPROGFILTBANK	Provides the allowable filter configurations
ITPSDFINAL	Provides the processed power spectral density information
J1939STATUS	Provides the status of the J1939 node
MODELFEATURES	Clearly states the features available for the current loaded model
RANGECMP4	Highly compressed version of the RANGE log

OEM7	Description
RAWSBASFRAME2	Provides the raw SBAS frame data and the transmitted frequency.
RTKASSISTSTATUS	Provides information on the state of RTK ASSIST
SAFEMODESTATUS	Provides the Safe Mode status

2.4 OEM6 to OEM7 – SPAN

The commands used to enter translational offsets and rotational offsets have changed in OEM7. For information about translations and rotations, see SPAN Translations and Rotations in the [OEM7 SPAN Installation and Operation User Manual](#).

2.4.1 Translational offsets

Determining the translational offsets in OEM7 is similar to the process in OEM6. Three dimensional distances from IMU to the other SPAN system device (e.g. GNSS antenna) are entered into the SPAN receiver. The differences are the commands used to enter the offsets have changed and the distances can be entered in the IMU frame or the vehicle frame.

See *Table 7: Commands for entering SPAN offsets and rotations* below for the changes in the commands used for entering translational offsets. See the **SETINSTRANSULATION** command for more information.

The OEM6 offset commands required that all translational offsets be entered in the IMU Body Frame, so to translate to the new commands, the same values can be entered within the new structure.

2.4.2 Rotational offsets

Determining rotational offsets in OEM7 has changed significantly. With the OEM6 commands, rotations usually referenced the SPAN Computation Frame. Now, all rotations are given as the rotation from the IMU Body Frame, to the frame of interest. For IMUs installed with their Z axis pointed upwards, rotation angles will be similar as those used with the legacy commands. For IMUs installed in other orientations, angles will be very different.

See *Table 7: Commands for entering SPAN offsets and rotations* below for the changes in the commands used for entering rotational offsets. See the **SETINSROTATION** command for more information.

Table 7: Commands for entering SPAN offsets and rotations

OEM6 command	OEM7 command	Notes
SETIMUTOANTOFFSET	SETINSTRANSULATION ANT1	
SETIMUTOANTOFFSET2	SETINSTRANSULATION ANT2	
SETIMUTOEXTOFFSET	SETINSTRANSULATION EXTERNAL	
SETIMUTOGIMBALOFFSET	SETINSTRANSULATION GIMBAL	
SETINSOFFSET	SETINSTRANSULATION USER	
SETMARKxOFFSET	SETINSTRANSULATION MARKx SETINSROTATION MARKx	Translational and rotational offsets are now entered separately. Markx can be MARK1, MARK2, MARK3 or MARK4.

OEM6 command	OEM7 command	Notes
VEHICLEBODYROTATION SETIMUORIENTATION	SETINSROTATION RBV	The information provided by the OEM6 commands is now combined in the RBV rotation.
GIMBALSPANROTATION SETGIMBALORIENTATION	SETINSROTATION RBM	The information provided by the OEM6 commands is now combined in the RBM rotation. The RBV rotation must also be specified for correct operation.
EXTHDGOFFSET	SETINSROTATION ALIGN	This offset continues to be automatically calculated if the translational offsets to the primary and secondary GNSS antennas are provided.
APPLYVEHICLEBODYROTATION	SETINSROTATION RBV	Default attitude output is now in the Vehicle Frame, as long as an RBV rotation is provided. This is overridden if a USER rotational offset is provided.

2.4.3 INS profiles

INS Profiles are a method to improve the performance of the system in different conditions. When a receiver is configured with an INS Profile, the default receiver settings are modified to optimize performance in the selected condition. See *INS Profiles* on page 28 for more information.

2.4.4 INS seed

The INS Seed functionality is an alignment method whereby INS alignment information from a previous powerup can be injected into the system at startup to achieve an INS alignment very quickly. This is especially useful for systems that previously required a kinematic alignment. See *INS Seed / Fast INS initialization* on page 31 for more information.

2.4.5 ALIGN calibration

For optimal SPAN performance when using Dual Antenna with SPAN, an ALIGN offset calibration is required for each unique installation. This calibration refines the IMU to antenna baseline angular offset from the initial estimate derived from the input lever arms. Refer to INSCALIBRATE command for details.

2.4.6 Multi-line body to vehicle calibration

The Body to Vehicle frame offset calibration feature assumes flat ground when estimating the roll offsets between the IMU Body and Vehicle frames. For applications where vehicle roll must be well known, or pass-to-pass accuracy of a highly offset position is especially important, the Multi-Line IMU Body to Vehicle frame offset calibration routine offers higher calibrated accuracy in roll. This is done by running independent calibrations over the same ground path in opposite directions; averaging the results cancels out the effect of any terrain induced roll. See *Multi-line body to vehicle frame rotation calibration routine* on page 29 for more information.

2.4.7 IMU-KVH1750 and IMU-P1750 baud rate conversion

If purchased directly from KVH, an IMU-KVH1750 or IMU-P1750 IMU must be configured before it can be used with an OEM7 receiver.

The IMU-KVH1750 and IMU-P1750 natively communicate at 921,600 bps, however the maximum baud rate of the COM ports on an OEM7 receiver is 460,800 bps. Thus, IMU-KVH1750 and IMU-P1750 IMUs purchased from KVH must be modified to use 460,800 bps. Also, the MSYNC setting on the IMU-KVH1750 or IMU-P1750 must be set to EXT.



Though slower, 460,800 bps is more than adequate to transmit the data at the required 200 Hz without any impact to performance.

There are two methods to change the IMU-KVH1750 or IMU-P1750 IMU baud rate. If an OEM6 receiver is available, a command is available that will allow the receiver to change the IMU baud rate. If not, a direct RS-422 connection at 921,600 baud is required to change the IMU baud rate.

Connect directly to the IMU



This method requires a direct connection to the IMU at 921,600 baud over RS-422. Make sure the RS-422 link used is capable of 921,600 baud as not all serial ports are capable of that rate.

1. Open a command terminal at 921,600 baud via an RS-422 serial link.
By default, the IMU will output 0.5 Hz binary messages which will help confirm you are connected properly.
2. Issue the following commands in sequence to enter configuration mode and change the baud rate.

```
=CONFIG,1
=MSYNC,EXT
=BAUD,460800
```
3. Upon completion of these commands, power cycle the IMU to boot up in the new baud rate.
4. Confirm the change by connecting to the IMU at 460,800 baud and verifying the incoming messages.

The new baud rate configuration is saved to the IMU NVM automatically so the process is complete and the IMU is ready for use with an OEM7 receiver.

Use an OEM6 receiver command

OEM6 firmware as of version OEM060631RN0000 (Dec 2016) includes the **IMUCONFIGURATION** command that internally configures the IMU-KVH1750 or IMU-P1750 IMU for SPAN communication. Three sets of configuration options are available, two of which allow changing the baud rate of the IMU to either 921,600 or 460,800. This provides customers upgrading to OEM7 an easy method to modify the baud rate. To change the IMU to 460800 baud, use the following procedure:

1. Connect the IMU to an RS-422 capable OEM6 receiver communication port.
2. Issue the following command.

```
CONNECTIMU COM# IMU_KVH_1750
```

3. Issue the following command.

```
IMUCONFIGURATION IMU_KVH_1750 2
```

The **IMUCONFIGURATION** command can be used to configure an IMU-KVH1750 or IMU-P1750 IMU depending on the value of the Option field.

```
IMUCONFIGURATION IMUType [Option]
```

IMU type	Option	Configuration details
IMU_KVH_1750	0	Configure KVH options for NovAtel communication.
	1	Change KVH baud rate to 921,600
	2	Change KVH baud rate to 460,800

2.4.8 INS Profiles

INS Profiles provide two major functions; simplified configuration and enhanced performance. Each INS Profile sets specific filter behavior to enhance system performance in particular environments.

The INS profile settings are enabled by the receiver model. A receiver model with an INS Mode of **S**, **P**, **R** or **K** is required to enable INS Profiles. The INS profile adds enhanced behavior such as Dead Reckoning for land and Heave for marine and allows for simple configuration rather than sending many discreet commands.

Table 8: OEM7 INS Profiles

Profile	Description	Enhanced Profile settings
DEFAULT	The default SPAN profile. This profile has legacy SPAN default settings.	N/A
LAND	INS Profile for fixed axle land vehicles. Enables robust kinematic alignment routines and configures the update profile.	Enables intelligent vehicle dynamics modeling for land vehicles (Dead Reckoning) Enables course over ground attitude updates Enables direction detection on kinematic alignment routine
MARINE	INS profile for marine vessels. Enacts changes to disable static behavior and make the marine solution more robust.	Disable Zero Velocity Updates Disable turn-on bias estimation Enables Heave algorithms
FIXEDWING	INS profile for fixed wing aircraft	None
FOOT	INS profile for walking or backpack applications	None
VTOL	INS profile for Vertical Take Off and Landing aircraft (helicopter, quadcopter, etc.)	None

Profile	Description	Enhanced Profile settings
RAIL	INS profile for railway applications	Enables intelligent vehicle dynamics modeling for rail vehicles (Dead Reckoning) Enable course over ground attitude updates Enable direction detection on kinematic alignment routine
AGRICULTURE	INS profile for agriculture applications	Enables direction detection on kinematic alignment routine Enables course over ground attitude updates Enables intelligent vehicle dynamics modeling for agricultural vehicles (Dead Reckoning)

Use

To use an INS Profile, send the **SETINSPROFILE** command at startup (or save to NVM) to activate the mode. For example:

```
SETINSPROFILE LAND
```

The type of profile activated, basic or enhanced, is determined by the SPAN model.



The currently selected profile option is available in the **INSCONFIG** log.

RBV calibration

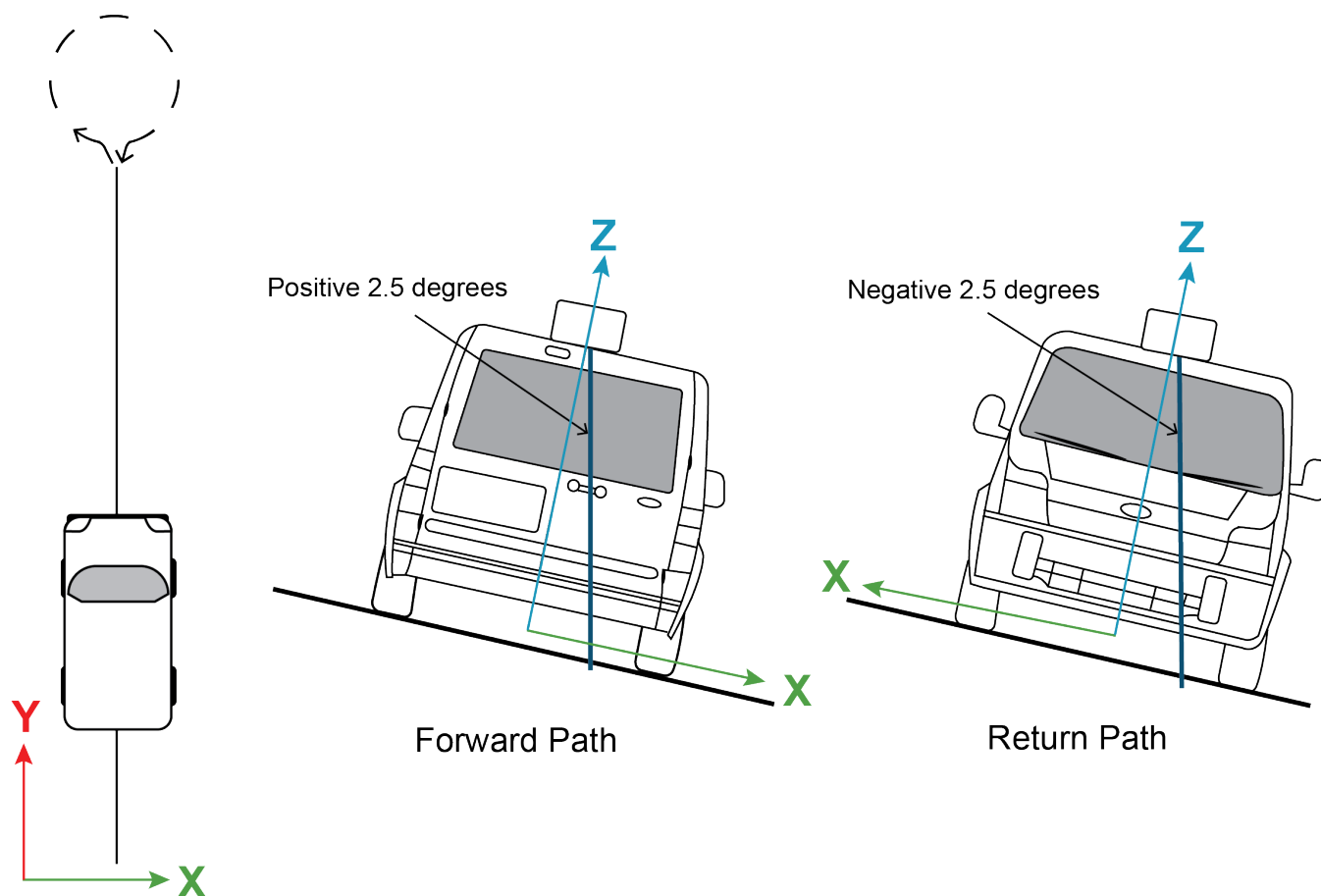
For the profiles to provide the greatest benefit, the rotation offset of the IMU body to the vehicle frame (RBV) should be known as accurately as possible. Since constraints happen at the vehicle level, any error in the offset will translate into worse performance. These parameters are estimated in the background when the system is fully converged, however it is always better to have this measured as precisely as possible before mission critical data is collected.

The INSCALIBRATE RBV functionality can be used to estimate and save these offsets to NVM in a controlled manner.

2.4.9 Multi-line body to vehicle frame rotation calibration routine

The single pass Body to Vehicle frame offset calibration feature assumes flat ground when estimating the roll offsets between the IMU Body and Vehicle frames, as this component of the rotational offset is difficult to observe through typical ground vehicle motion. In practice this can result in an outstanding roll offset error as high as several degrees. For applications where vehicle roll must be well known, or pass-to-pass accuracy of a highly offset position is especially important, the Multi-Line IMU Body to Vehicle frame offset calibration routine offers higher calibrated accuracy in roll. This is done by running independent calibrations over the same ground path in opposite directions; averaging the results cancels out the effect of any terrain induced roll. A minimum of two iterations are required. Additional iteration pairs can be executed for increased reliability.

Figure 1: Multi-line IMU body to vehicle calibration



The steps for the calibration routine are:

1. Apply power to the receiver and IMU.
2. Configure the SPAN system.
3. Ensure that an accurate lever arm has been entered into the system.
4. Perform an initial system alignment using one of the methods described in System Start-Up and Alignment Techniques in the [OEM7 SPAN Installation and Operation User Manual](#). Ensure the system reaches Solution Good.
5. Enable a new calibration using the **INSCALIBRATE** command, with the optional standard deviation target omitted:

```
INSCALIBRATE RBV NEW
```

6. Start to move the vehicle. Drive a straight course, at a vehicle speed greater than 5 m/s (18 km/hr). A distance of approximately 250 metres is recommended.

To monitor the calibration, log **INSCALSTATUS** using the ONCHANGED trigger.

7. When the end of the available distance is reached, stop the vehicle. Halt the ongoing calibration by sending the **INSCALIBRATE** command again:

```
INSCALIBRATE RBV STOP
```

8. At this point **INSCALSTATUS** will report a source status of CALIBRATED. Turn the vehicle around and come to a stop on the finishing point of the previous line.

9. Initiate the second pass of the calibration by issuing the **INSCALIBRATE** command with the **ADD** parameter:

```
INSCALIBRATE RBV ADD
```

10. Drive the vehicle back along the original course, until you reach the starting point for the first pass. Stop the vehicle, and re-issue the **INSCALIBRATE** command, with the **STOP** parameter. **INSCALSTATUS** will again indicate a source status of CALIBRATED.

At this point the Multi-Line calibration can be considered complete, but more iterations can be repeated until the desired accuracy level is reached.

At any point during an active calibration line, the **INSCALIBRATE** command, with the **RESET** parameter can be used to reset the calibration. This will ensure that the results of the active line are not added to the Multi-Line calibration. The **INSCALSTATUS** log will restore the previous source status and calibration values.

11. When the calibration is complete, the calibrated offset value can be viewed in either the **INSCALSTATUS** log or the **INSCONFIG** log.

To save a calibrated rotation for subsequent start ups, issue the **SAVECONFIG** command after calibration is complete. Each time the IMU is re-mounted this calibration should be performed again.



Do not start INS Calibrations (NEW or ADD) until the inertial alignment is completed and system has converged.



For information about the logs and commands used in this procedure, refer to the [OEM7 Commands and Logs Reference Manual](#).

2.4.10 INS Seed / Fast INS initialization

The INS Seed functionality is an alignment method whereby INS alignment information from a previous powerup can be injected into the system at startup to achieve an INS alignment more quickly. This is especially useful for systems that previously required a kinematic alignment.

When INS Seed is enabled, alignment data is automatically saved when appropriate, with no specific action required from the user. During subsequent power ups, the saved solution is compared to the current position of the system, and if valid, the saved solution is used to immediately initialize the inertial filter, making it ready for use.

Error model information is also saved with the seed data. Even if full alignment cannot be achieved, this information will still be used to aid filter convergence, although another method of alignment will be required.

This feature is enabled via the **INSSEED** command. This command setting must be saved to NVM via the **SAVECONFIG** command to be used upon next system boot-up.



When the INS Seed functionality is enabled, it will most often be the first available alignment method and therefore the used alignment method. If Dual Antenna Alignment is desired, consider disabling INSSEED.

Criteria

For the INS seed functionality to be enabled, the **INSSEED ENABLE** command must be sent.

Saving

The following criteria must be met for valid alignment data to be saved into receiver NVM.

- The INS solution status must be converged, as indicated by the INS solution convergence flag in the inertial Extended Solution Status.
- The system velocity is less than 0.2 m/s.

When these conditions are met, the required information is automatically saved to NVM. If these conditions cease to be met, the saved alignment data is automatically discarded. If the alignment data is discarded, error model information is automatically retained as appropriate.

Use at boot-up

Upon boot, several checks are done to verify that any seed data existing in NVM is appropriate for use. These checks require a GNSS position to be computed, so an NVM seed will never be injected prior to achieving a GNSS position. The following criteria must be met in order to use a seed:

- Seed data from NVM must be valid
- The system must be stationary
- The system must not have moved significantly from the seed position: less than 10 metres in position and 10 degrees in heading

If a valid error model is present in the seed data from NVM, this will always be used, even if the system is not stationary or exceeded the movement thresholds.

INJECT Option (Advanced Users Only)

There is an advanced option available to skip the second and third validation steps described in the boot-up section above. This can be used if GNSS is not available on power-up or if speed to achieve an alignment is imperative.

This option is for advanced users only. Forcing an injection of an NVM seed without the validation checks can cause an unstable INS solution if the vehicle has moved.

Apart from injecting the seed at start-up without validation, this will also remove some requirements typically asserted to allow the filter to converge. This will allow the system to run without GNSS adequately at start-up.

Example usage

Enabling functionality after FRESET:

1. Send the following command.

```
INSSEED ENABLE VALIDATE
```

2. Send the following command.

```
SAVECONFIG
```

Saving valid INS information:

1. Operate the system as normal, until the filter completes convergence.
2. Bring the system to a complete stop before powering off or resetting.

Using valid INS information on start-up:

1. Remain static while initializing.
2. Wait for the system to receive a valid position from GNSS.
3. If valid, alignment data will be injected.

System indicators

The **INSSEEDSTATUS** log reports the seed injection status for the current power up and information on current seed validity. Refer the [OEM7 Commands and Logs Reference Manual](#) for more information about the **INSSEEDSTATUS** log.

The Extended Solution Status field in the **INSPVAX** log has bits that indicate if the alignment direction is verified, the Alignment type that occurred and NVM seed injection status.

Alignment direction verified: Bit 25

This bit indicates whether the alignment is verified or not. When the seed is injected and kinematic validation is finished, bit 25 is set to 1. Note that this bit will not be set until the kinematic validation completes. When the INJECT option is used, validation is skipped and this bit will not be set.

Alignment type: Bits 26-28

These bits indicate how the INS system reached INS_ALIGNMENT_COMPLETE. When the INS system aligns using saved INS solution data from NVM, bits 26-28 will be set to 101

NVM seed status: Bits 29-31

These bits indicate the current status of the seeding process at start-up, as described in the table below:

Table 9: NVM seed indication

Bit 31-29 values ¹	Hex value	NVM seed type
000	0x00	INS Seed has not been injected into the solution
001	0x01	Valid INS Seed was not found in non-volatile memory
010	0x02	INS Seed has failed validation and has been discarded
011	0x03	INS Seed is awaiting validation
100	0x04	INS Seed alignment data has successfully been injected (including error model data)
101	0x05	INS Seed exists, but has been ignored due to a user commanded filter reset/restart or configuration change
110	0x06	INS Seed error model data has successfully been injected

1. The values from left to right are Bit 31, Bit 30 and Bit 29.

Chapter 3 OEM6 to OEM7 – hardware

The following OEM7 GNSS receivers are discussed in the following sections.

- OEM719
- OEM729

3.1 Physical changes

Two of the OEM7 receivers are form factor replacements for OEM6 receivers. The OEM719 replaces the OEM615 and the OEM729 replaces the OEM628. Form factor replacement means that the OEM7 receivers:

- have the same physical dimensions as the OEM6 receivers
- have the same keep-out zones as the OEM6 receivers
- use the same locations for the mounting holes as the OEM6 receivers
- use the same connectors as the OEM6 receivers
- use the same locations for the connectors as the OEM6 receivers

The physical differences between the OEM7 and OEM6 receivers are described in the following sections.

3.1.1 Shielding

The OEM719 and OEM729 have an additional shield compared to the OEM615 and OEM628. However, all of the shields on the OEM719 are within the keep-out zone defined for the OEM615 and all of the shields on the OEM729 are within the keep-out zone defined for the OEM628. Designs that respected the keep-out zones for the OEM6 receivers will not be affected by the shields on the OEM719 and OEM729.

For information about the OEM615 and OEM628 keep-out zones, see the [OEM6 Family Installation and Operation User Manual](#) (OM-20000128). For information about the OEM719 and OEM729, see the [OEM7 Installation and Operation User Manual](#).

3.1.2 Mounting

The OEM719 and OEM729 can be mounted using standoffs or bosses that have the same spacing required for OEM6 receivers. See the [OEM7 Installation and Operation User Manual](#) for the exact spacing.

OEM7 receivers also have an area on the edge of the cards to allow the use of mounting rails. This is the area outside of the keep-out zone. On the OEM719, this area is 1.75 mm wide. On the OEM729, this area is 2.5 mm wide.

Using rails to mount the OEM7 receiver improves the thermal and vibration performance of the receiver. Securing the OEM7 receiver to mounting rails using clamping bars provides the most secure configuration for aggressive thermal and vibration use cases.



For more details about mounting OEM7 receiver cards, refer to the *OEM7 Receiver Card Mechanical Integration Application Note* (D19021).

3.1.3 Weight

The OEM7 receivers are slightly heavier than the OEM6 receivers they replace.

Receiver	Weight		Receiver	Weight
OEM615	24 g		OEM628	37 g
OEM719	31 g		OEM729	48 g

3.2 Electrical changes

3.2.1 COM port data rate

The maximum data rate for COM ports has changed. On OEM6 receivers, the maximum data rate was 921,600 bps. On OEM7 receivers, the maximum data rate is 460,800 bps.

3.2.2 Power supply

The power consumption of OEM7 receivers is dependent on receiver configuration and feature use. Enabling more features (e.g. interference detection) will increase the power required.

- The OEM719 power consumption is approximately 10% higher than a similarly configured OEM615.
- The OEM729 power consumption is approximately 10% lower than a similarly configured OEM628.

A monotonic rise of the supply voltage input is required to guarantee a proper power-on reset sequence. The maximum rise time is 100 ms.

Bulk supply bypassing (approx 10 μ F) and high-frequency bypassing (0.1 μ F and 220 pF) near the supply pin is recommended for optimal performance.

On the OEM7 receiver cards, there is an approximate effective input bulk capacitance of 100 μ F at room temperature. This is made up of ceramic capacitors with a voltage rating of 6.3 VDC and tolerance of 20%.

OEM7 receivers monitor the supply voltage. If the supply voltage falls outside of the normal operating range, a warning or error is provided in the RXSTATUS log. See the **RXSTATUS** log for more information.

Inrush current

The inrush current behavior of the OEM7 receivers is different than the OEM6 receivers.

Table 10: OEM6 receiver inrush current

OEM6 receiver	Inrush current (typical)
OEM615	6.0 A for less than 60 μ s
OEM617	6.0 A for less than 60 μ s
OEM617D	6.0 A for less than 60 μ s
OEM628	6.6 A for less than 60 μ s
OEM638	12.0 A for less than 60 μ s

Table 11: OEM7 receiver inrush current

OEM7 receiver	Inrush current (typical)
OEM719	1.71 A for less than 1.5 ms

OEM7 receiver	Inrush current (typical)
OEM729	2.0 A for less than 1.8 ms
OEM7500	1.7 A for less than 1.5 ms (3.3 VDC) No inrush current for 1.2 VDC
OEM7600	2.0 A for less than 1.8 ms
OEM7700	2.0 A for less than 1.8 ms
OEM7720	2.0 A for less than 1.8 ms

3.2.3 USB interfaces

The layout guidelines are identical to those provided for OEM6, but the recommended common-mode choke value has changed.

The OEM719 and OEM729 USB interfaces operate at 12 Mb/s ("Full-Speed") operation for backwards-compatibility reasons.

3.2.4 Input and output lines

The voltage limits and drive currents on the input and output lines of the OEM7 receivers are different than the levels on the OEM6 receivers.

3.2.5 LNA power

If a short circuit or other problem causes an overload of the current supplied to the antenna, the receiver hardware shuts down the power supplied to the antenna. To restore power, power cycle the receiver. The Receiver Status word, available in the RXSTATUS log, provides more information about the cause of the problem.

OEM719

On OEM615 receivers, LNA power is provided by an external power supply connected to pin 1 of the main connector.

On OEM719 receivers, LNA power is generated on the receiver and pin 1 of the main connector has no electrical connection on the board. The OEM719 provides up to 200 mA of current at +5 VDC. The LNA is generated from the 3.3 V supply; therefore the OEM719 will draw more power on the 3.3 V rail than a similarly configured OEM615.

The typical current draw for a NovAtel active antenna is <40 mA. For a typical antenna, this may increase the power drawn from the 3.3 V supply by as much as 70 mA (as the LNA supply is approximately 85-90% efficient).

OEM729

On OEM628 receivers, the voltage provided to the LNA could be changed from the default of +5 VDC to +3.3 VDC using the ANTENNAPOWER command.

The voltage for the LNA power is not selectable on OEM729 receivers, but the OEM729 can provide more current to the LNA. The OEM729 provides up to 200 mA of current at +5 VDC.

3.3 Environmental changes

3.3.1 Thermal dissipation

To ensure functionality and reliability, the OEM7 receiver cards must operate within the specified ambient air temperature limits (-40°C to +85°C).

The OEM7 receivers have been designed to efficiently transfer heat from the receiver components into the printed circuit board. Mounting the OEM7 receiver on rails, or attaching heat sinks to the mounting areas, will transfer the heat from the receiver card to adjacent circuit boards, the enclosure or the air.



The mounting area is the area on the sides of the receiver that are outside of the keep-out zone.



The best thermal mitigation comes from sinking heat through the mounting rails. This provides more effective thermal relief than attaching a heat sink to the MINOS.



For more information about thermal dissipation, refer to the *OEM7 Receiver Card Mechanical Integration Application Note* (D19021).

3.3.2 Vibration

On OEM6 receivers, the random vibration specification is dependent on the receiver variant. Standard OEM6 receivers were rated to 7.7g RMS (MIL-STD-810G Method 514.6, Category 24). High vibration OEM6 receivers were rated to 20g RMS (MIL-STD-810G Method 514.6, Category 24).

All OEM7 receivers are rated to 20g RMS (MIL-STD-810G Method 514.6E-1, Category 24). However, for high vibration installations, special considerations are required.



For more information about vibration considerations, refer to the *OEM7 Receiver Card Mechanical Integration Application Note* (D19021).

3.4 Interference

Typical GNSS signal levels are in the order of -130 dBm. Spurious emissions from digital logic (especially a microprocessor) or a nearby radio transmitter can easily meet the appropriate EU or FCC emission limits and still be several orders of magnitude above the level of the GNSS signals.

These emissions may desensitize the GNSS receiver and lead to performance degradation (low carrier to noise ratios, position accuracy, or loss of lock in extreme cases).

There are two bands of interest:

- Low Band (1164 MHz to 1300 MHz): contains GPS L2, GPS L5, GLONASS L2, BeiDou B2, BeiDou B3, Galileo E5a, Galileo E5b, Galileo E5 and Galileo E6.
- High Band (1525 MHz to 1610 MHz): contains GPS L1, GLONASS L1, BeiDou B1, Galileo E1, and L-Band.

OEM7 receivers are designed to mitigate RF interference on the GNSS card itself. However, if an emission (fundamental or harmonic) from equipment co-located with the GNSS receiver or antenna falls into one of these bands, there are a few things that can be done to ensure good system performance:

- Place radio transmitters and their associated antennas as far away from the GNSS antenna and OEM7 card as possible (for example, a cellular radio antenna).
- Provision for RF shielding over all high-speed digital logic circuitry (even if the design meets the required emissions regulations). A CPU operating at 400MHz will produce an emission at 1200MHz (3rd harmonic) and another at 1600MHz (4th harmonic) – the emissions may be strong enough to affect the GNSS system performance. Similar provisions should be made for any RF circuitry near the GNSS receiver or antenna.
- Run the OEM7 antenna coaxial cable away from high-speed digital logic, other radio sections and high-current nets (as in switching power supplies).

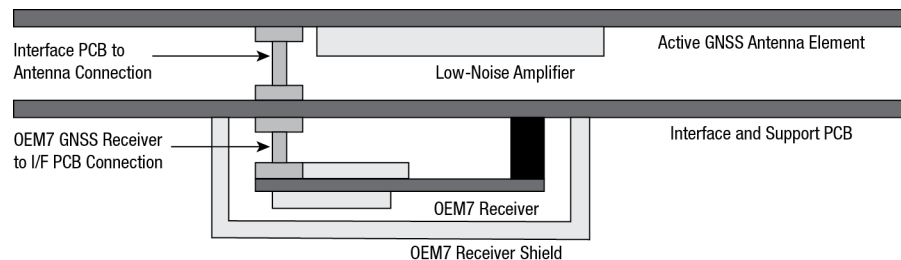
3.4.1 Interference from co-located LNA

Often a GNSS receiver is placed in the same enclosure as a GNSS antenna and LNA to create an all in one GNSS receiver product (also referred to as a *smart antenna*).

The close proximity of a high-gain amplifier to the GNSS receiver may cause some low-level near-field energy from the receiver to be amplified and fed back into the receiver, causing C/No degradation to one or more constellations.

It is strongly advised in a smart antenna design that the receiver be enclosed in a shield to prevent this near-field energy from coupling into the antenna. This includes connecting the GNSS receiver input ground to the shield surrounding the receiver, similar to the following image. It is also strongly advised to use mounting rail for receiver cards used in this configuration.

Figure 2: SMART antenna interference example



The previous figure shows one method of shielding the receiver.



The rail mounting provision on the edges of the receiver card can also be used to attach entire board shielding.

3.1 Receiver card interface examples

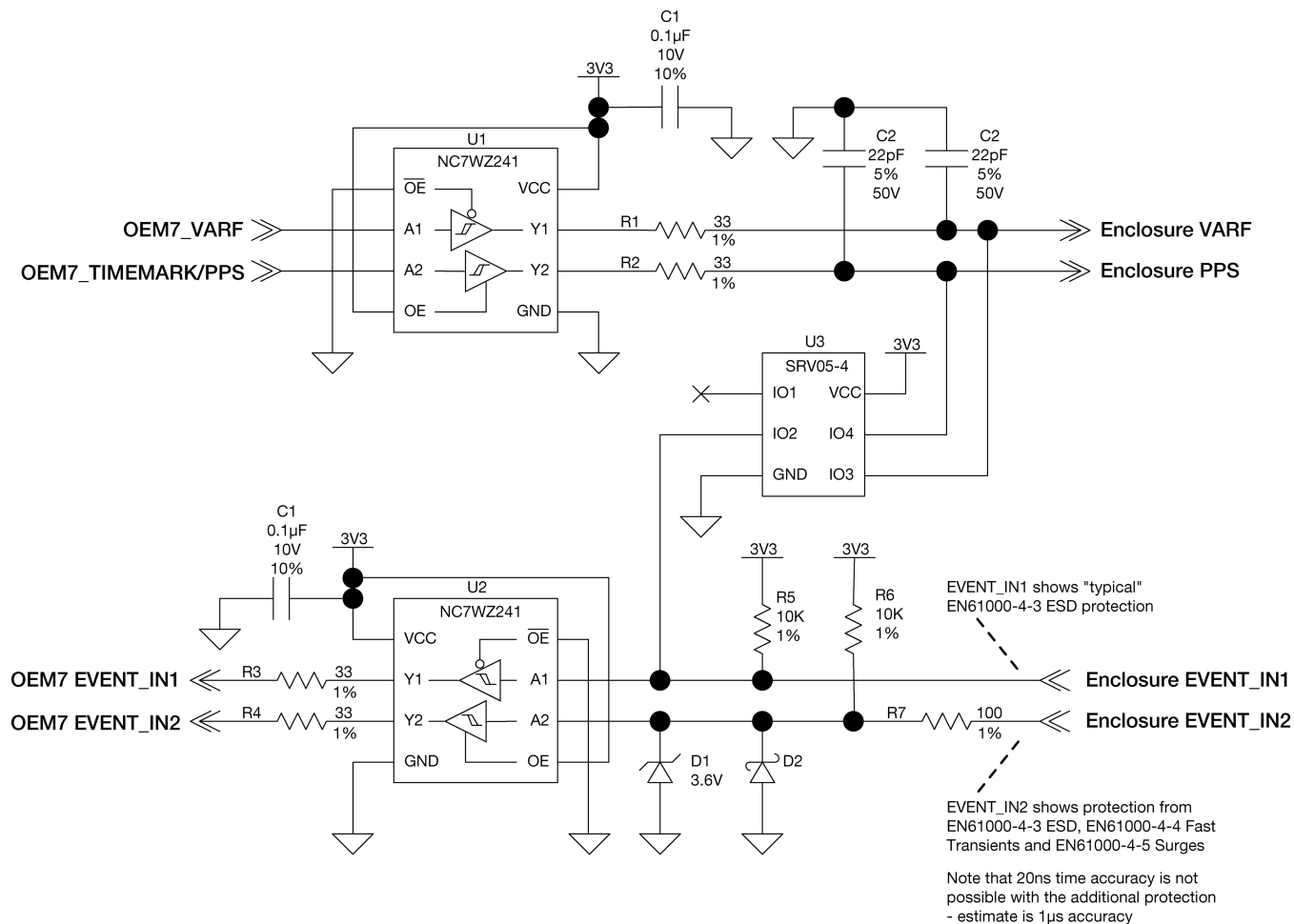
The OEM7 receiver cards provide a number of 3.3V CMOS-level I/O pins for status indication and timing:

- **COMx**: CMOS-level UART ports
- **Ethernet**: 10/100 Ethernet port
- **CAN1** and **CAN2**: CMOS-level CAN ports (external CAN transceivers required)
- **PPS**: Output pulse providing time reference signal (software configurable output rate, defaults to 1 Pulse Per Second)
- **EVENTx_OUT**: Variable Frequency output (a software-configurable clock output similar to PPS, off by default). (The EVENT1_OUT signal may also be referred to as an VARF.)
- **PV**: Position Valid (This signal drives high when the receiver has calculated a valid position)
- **EVENTx_IN**: Event inputs (with configurable polarity)
- **USERIO**: A user defined general purpose input or output (OEM729 only)

These I/O require additional ESD protection if they are to be routed to enclosure connectors. A protection circuit similar to the following examples must be used on any OEM7 CMOS-level signals that will be routed to an enclosure connector. The ferrite bead and small-value capacitor provide some immunity to electrostatic discharge events, but also serve to reduce radiated and conducted emissions from the enclosure.

3.4.2 *EVENT_IN, EVENT_OUT and PPS signal protection*

The following schematic shows appropriate signal conditioning for EVENT_IN (MKI), EVENT_OUT (VARF) and PPS (TIMEMARK) signals.

Figure 3: Protection and buffering for EVENT_IN, PPS and EVENT_OUT signals

The buffers (U1, U2) are chosen to provide additional drive strength for long cables. The series terminations (R1, R2, R3, R4) are present to allow for signal integrity adjustments (to reduce overshoot/undershoot, etc.). The passive component values may require tuning, depending on the application.

The TVS (U3) provides protection from ESD/EFT events and some induced power line surges. There are two implementations shown for the EVENT_IN signals. EVENT_IN1 is shown with basic ESD/EFT protection, while EVENT_IN2 is shown with additional surge protection for harsh environments (see D1, R7, D2, D3 and R9). If the additional surge protection is to be used on a design, the buffer must be included as well. This design has been used to protect EVENT_IN signals from EN61000-4-5 induced surges up to 2kV on several NovAtel enclosure products. If nanosecond-level timing is critical to the application, this protection circuit may be unsuitable, as it causes a timing shift of a few microseconds to the EVENT signal (heavily dependent on temperature).

The 10 kΩ pullups (R5, R6) are required if the buffers are used on the EVENT_IN lines. If the buffers are not used, the OEM7 cards have built-in 10 kΩ pullup resistors – external pull resistors are not required.

Table 12: EVENT_IN, EVENT_OUT and PPS pin designation

	OEM719	OEM729	OEM7600	OEM7700	OEM7720
Connector	P1701	P1802	P1701	P2001	P1901

	OEM719	OEM729	OEM7600	OEM7700	OEM7720
EVENT_IN1	9 ¹	8	30	30	30
EVENT_IN2	7 ²	7 ³	29	29	29
EVENT_IN3	–	–	32	32	32
EVENT_IN4	–	–	31	31	31
PPS	19	4	24	24	24
EVENT_OUT1/VARF	6 ⁴	3	17	17	17
EVENT_OUT2	–	–	26	26	26
EVENT_OUT3	–	–	23	23	23
EVENT_OUT4	–	–	25	25	25

Table 13: Bill of materials (critical components)

Designator	Manufacturer	Manufacturer part number
D1	On Semiconductor	MMSZ5227BT1G
D2	Comchip Technology	CDBU00340
U1, U2	Fairchild Semiconductor	NC7WZ241L8X
U3	Semtech	SRV05-4A.TCT

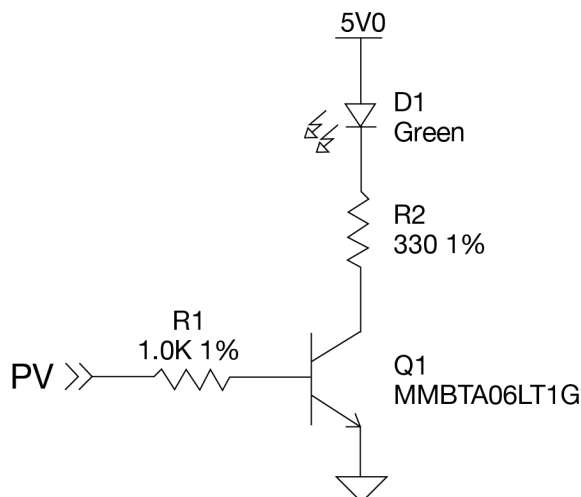
3.4.3 Position Valid (PV) LED driver

The PV signal may be used to indicate that the receiver card has computed a valid position. Many NovAtel enclosure products use it to drive a green LED on the enclosure. It may also be used to monitor the status of the receiver with an external microcontroller.

If the LED requires less than 6 mA to drive, and can be operated from a 3.3 V supply, the PV pin may be used to drive the LED directly (with an appropriate current-limiting resistor). If the LED must be driven from a different supply voltage or requires more than 6 mA, a buffer must be used.

A simple buffer – suitable for driving a high-brightness LED from a 5 V supply – is shown below. The value for R2 was selected to limit the LED current to 10 mA (assuming $V_{CE(Q1,sat)}=0.25\text{ V}$, $V_{FWD(D1)}=3.4\text{ V}$). This circuit will not compensate for the forward voltage change over temperature (and the resulting brightness changes).

-
1. This pin is multiplexed with the COM3 transmit signal and is not available if COM3 is enabled.
 2. This pin is multiplexed with the CAN1 transmit signal and is not available if CAN1 is enabled.
 3. This pin is multiplexed with the COM3 receive signal and is not available when COM3 is enabled.
 4. This pin is multiplexed with the CAN1 receive signal and is not available if CAN1 is enabled.

Figure 4: OEM7 buffer for driving high-brightness LEDs from PV**Table 14: PV pin designation**

	OEM719	OEM729	OEM7600	OEM7700	OEM7720
Connector	P1701	P1802	P1701	P2001	P1901
PV	17	10	22	22	22

Table 15: PV LED driver bill of materials (critical components)

Designator	Manufacturer	Manufacturer part number
Q1	ON Semiconductor	MMBTA06LT3G
	Fairchild Semiconductor	MMBTA06LT1G
	Diodes Inc	MMBTA06-7-F
D1	OSRAM	LT L29S-N1R2-25-Z

3.4.4 Communication ports

The COM ports on OEM7 receivers (except COM1 on the OEM729) are CMOS-level I/O pins only. These ports require the addition of an RS-232 or RS-422 transceiver to provide appropriate signal levels. Most RS-232 or RS-422 transceivers will provide some ESD protection (for harsh environments, additional protection may be required).



COM1 on the OEM729 uses RS-232 protocol by default. This port can be changed to RS-422 protocol using the **SERIALPROTOCOL** command.

A combination of a series ferrite bead and small-value shunt capacitor is recommended on any RS-232/RS-422 lines that leave the enclosure (similar to the arrangement used on the I/O ports and CAN interfaces). Many of the available transceivers provide protection for ESD events to $\pm 15\text{kV}$ – if additional protection is required, an external TVS with a working voltage of $\pm 15\text{V}$ to $\pm 25\text{V}$ will be required.

3.4.5 CAN controller ports

OEM7 receivers provide 3.3 V CMOS-level CAN controller ports. An external CAN transceiver is required. The following figure shows a typical CAN transceiver implementation.

The combination of ferrite beads and small-value capacitors are not necessarily required but may provide improved EMI performance. A low-capacitance TVS device is shown on the schematic to provide ESD protection.

Figure 5: OEM7 CAN transceiver example

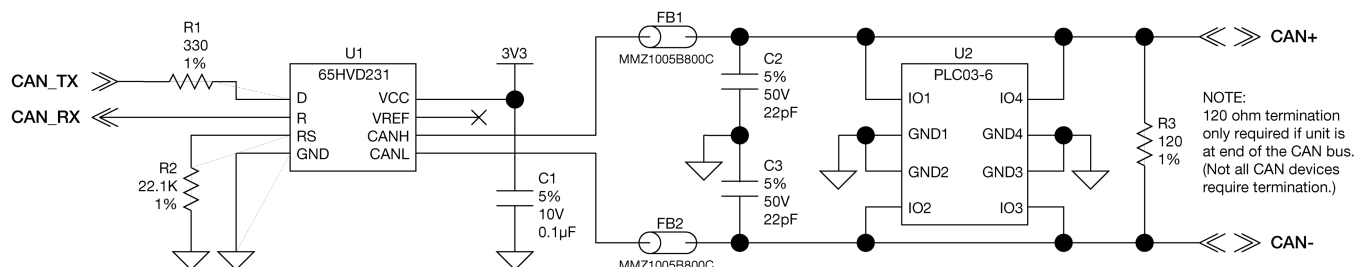


Table 16: CAN transceiver pin designation

	OEM719	OEM729	OEM7600	OEM7700	OEM7720
Connector	P1701 ¹	P1803	P1701	P2001	P1901
CAN1TX	7	10	36	36	36
CAN1RX	6	11	38	38	38
CAN2TX	20	12	37	37	37
CAN2RX	8	13	35	35	35

The 120Ω termination resistor should only be used when the CAN device is used at one end of the CAN bus. Multiple terminations along the length of the CAN bus will degrade performance for all CAN devices on that bus.

The slew rate adjustment resistor (R2) value shown sets the slew rate for applications for SAE J1939 agricultural applications. Other applications may require a different slew rate. Refer to the transceiver data sheet for more information.

Table 17: CAN transceiver example bill of materials (critical components)

Designator	Manufacturer	Manufacturer part number
FB1, FB2	TDK	MMZ1005B800C
U1	Texas Instruments	SN65HVD231QD
U2	Bourns	CDNBS08-PLC03-6

1. Pins 6 and 7 are multiplexed with other signals and may not be available when certain features are enabled.

3.4.6 USB interface



This section describes the interface requirements for an OEM7 receiver card USB port. For general information about the OEM7 receiver card USB ports.

OEM7 receivers include an USB 2.0 device interface set to either Full-Speed (12 Mb/s) or High-Speed (480 Mb/s) transfer rate. On the OEM719 and OEM729 receivers, the transfer rate for the USB Device port is Full-Speed (12 Mb/s). The USB device interface on the OEM7600, OEM7700 and OEM7720 receivers use the High-Speed (480 Mb/s) transfer rate. The device interface will not auto-negotiate the speed with the Host computer.

It is imperative for signal integrity and EMI reasons that the differential data traces be routed as a 90Ω differential pair. Use of a small-value common-mode choke (as shown in *Figure 6: OEM7 USB device interface example* below) may improve the radiated emissions performance (but should not be necessary).

Any stubs on the traces must be kept as short as possible and it is strongly recommended not to change reference planes. Match the USB differential pair (D+ and D- trace lengths) to within 3 mm or less.

The common-mode choke and ESD protection should be placed as close as possible to the USB connector (J1).

Figure 6: OEM7 USB device interface example

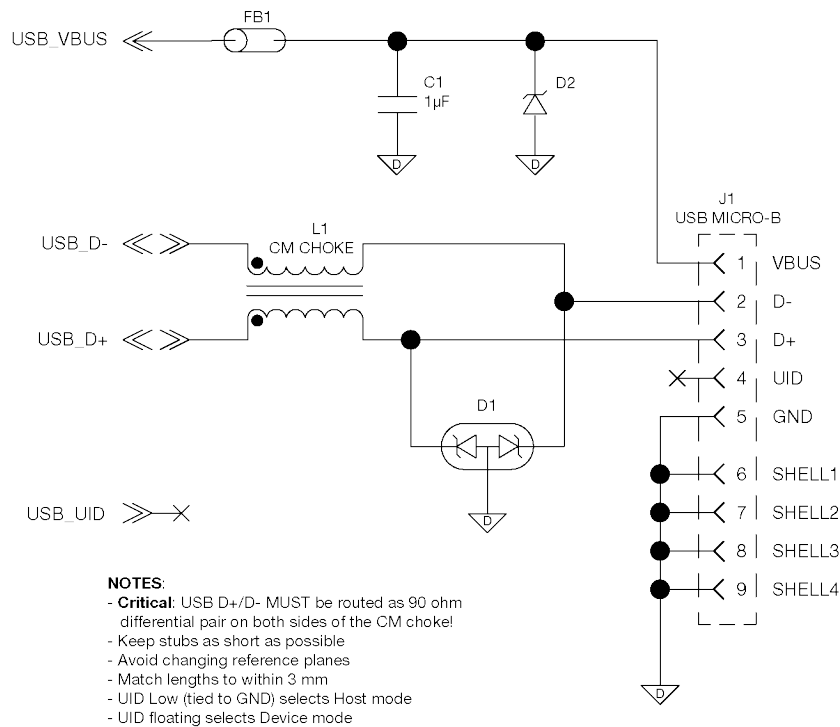


Table 18: USB device interface pin designation

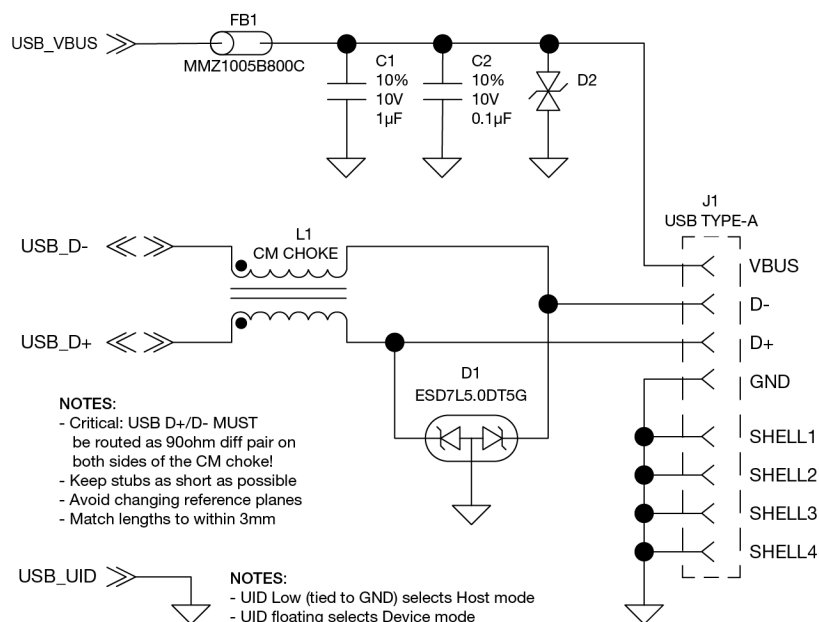
	OEM719	OEM729	OEM7600	OEM7700	OEM7720
Connector	P1701	P1802	P1803	P1701	P2001
USB0_D-	3	21	-	50	50

	OEM719	OEM729	OEM7600	OEM7700	OEM7720
USB0_D+	4 ¹	22	-	48	48
USB1_D-	-	-	-	47	47
USB1_D+	-	-	-	49	49
USB_VBUS	-	-	15	52	52
UID	-	-	14	51	51

Table 19: USB device interface example bill of materials

Designator	Manufacturer	Manufacturer part number
D1	On Semiconductor	ESD7L5.0DT5G
D2	On Semiconductor	ESD5Z6.0T1G
FB1	TDK	MMZ1005B800C
L1	Würth Electronics	744230900
J1	FCI	10104110-0001LF

Figure 7: OEM7 USB host interface example



1. This pin is multiplexed with the COM3 receive signal and is not available when COM3 is enabled.

Table 20: USB host interface pin designation

	OEM719	OEM729		OEM7600	OEM7700	OEM7720
Connector	P1701	P1802	P1803	P1701	P2001	P1901
USB0_D-	3	21	-	50	50	50
USB0_D+	4 ¹	22	-	48	48	48
USB1_D-	-	-	-	47	47	47
USB1_D+	-	-	-	49	49	49
USB_VBUS	-	-	15	52	52	52
UID	-	-	14	51	51	51

Table 21: USB host interface example bill of materials

Designator	Manufacturer	Manufacturer part number
D1	On Semiconductor	ESD7L5.0DT5G
D2	Littelfuse	V5.5MLA0402NR
FB1	TDK	MMZ1005B800C
J1	Molex	0482040001
L1	Würth Electronics	744230450

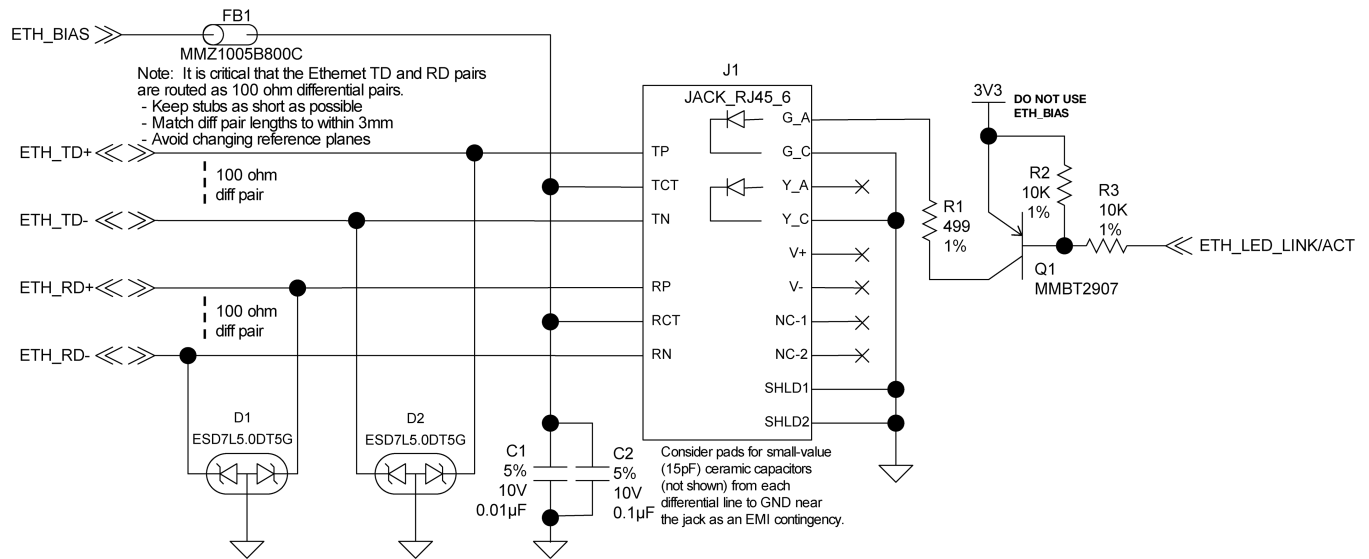
3.4.7 Ethernet port

OEM7 receivers, except the OEM719, provide a 10/100 Ethernet port with auto-negotiation. The Ethernet interface is disabled by default and must be configured. See the Ethernet Configuration OEM7 User Documentation Portal (docs.novatel.com/OEM7) for instructions on Ethernet device configuration. The PHY layer is based on the Texas Instruments TLK106 Ethernet PHY.

PHY terminations are provided on the OEM7 receiver card and a 3.3 V output is presented to bias the Ethernet magnetics. The 3.3 V power supplied by the OEM7 receiver card is not to be used for any purposes other than biasing the Ethernet magnetics.

A reference schematic is shown in *Figure 8: Ethernet reference schematic* on the next page. The ferrite bead is included as an EMI de-risk contingency and may not be necessary.

1. This pin is multiplexed with the COM3 receive signal and is not available when COM3 is enabled.

Figure 8: Ethernet reference schematic**Table 22: Ethernet pin designation**

	OEM729	OEM7600	OEM7700	OEM7720
Connector	P1803	P1701	P2001	P1901
ETH_BIAS	Receive: 3 Transmit: 6	56	56	56
ETH_TD+	4	58	58	58
ETH_TD-	5	60	60	60
ETH_RD+	2	57	57	57
ETH_RD-	1	59	59	59

Run 100 Ω differential pairs over unbroken reference planes directly to the jack. Ensure the integrated magnetics in the jack meet the specifications in *Table 23: Ethernet transformer characteristics* below. Ensure that the jack is no more than 15 cm (6 inches) from the OEM7 connector. Shorter runs are better.

Table 23: Ethernet transformer characteristics

Parameter	Value	Test condition
Turns ratio	1 CT : 1 CT	
Open-CCT inductance (minimum)	350 μ H	100 mV, 100 kHz, 8 mA
Leakage inductance (maximum)	0.4 μ H	1 MHz (minimum)
Inter-winding capacitance (minimum)	12 pF	
DC resistance (maximum)	0.9 Ω	
Insertion loss (maximum)	1.0 dB	0 MHz - 65 MHz
HIPOT (minimum)	1500 Vrms	

Table 24: Bill of materials (critical components)

Designator	Manufacturer	Manufacturer part number
FB1	TDK	MMZ1005B800C
D1, D2	On Semiconductor	ESD7L5.0DT5G
J1	Wurth Electronics	7499211121A
Q1	On Semiconductor	MMBT2907ALT1G
	Fairchild Semiconductor	MMBT2907A

3.5 Differences between PwrPak7 and OEM6 enclosures

The following tables compare the specifications and features of the PwrPak7 with the specifications and features of OEM6 enclosures.

Table 25: Physical specifications

	PwrPak7-E1 PwrPak7D-E1	PwrPak7 PwrPak7D	FlexPak6	FlexPak6D	ProPak6
Dimensions	147 x 125 x 55 mm	147 x 125 x 55 mm	147 x 113 x 45 mm	147 x 113 x 45 mm	190 x 185 x 75 mm
Weight	510 g	500 g	337 g	315 g	1.79 kg

Table 26: Power requirements

	PwrPak7 (all models)	FlexPak6	FlexPak6D	ProPak6
Input voltage	9 to 36 VDC	6 to 36 VDC	6 to 36 VDC	9 to 36 VDC
Power consumption	1.8 W	1.8 W	1.9 W	3.5 W

Table 27: Signals tracked

		PwrPak7 PwrPak7-E1	PwrPak7D PwrPak7D-E1	FlexPak6	FlexPak6D	ProPak6
GPS	L1 C/A	Y	Y	Y	Y	Y
	L1C	Y	Y	–	–	–
	L2C	Y	Y	Y	Y	Y
	L2P	Y	Y	Y	Y	Y
	L5	Y	Y	Y	–	Y

		PwrPak7 PwrPak7-E1	PwrPak7D PwrPak7D-E1	FlexPak6	FlexPak6D	ProPak6
GLONASS	L1 C/A	Y	Y	Y	Y	Y
	L2 C/A	Y	Y	Y	Y	Y
	L2P	Y	Y	Y	–	Y
	L3	Y	Y	–	–	–
Galileo	E1	Y	Y	Y	Y	Y
	E5a	Y	Y	Y	–	Y
	E5b	Y	Y	Y	Y	Y
	E5 AltBOC	Y	Y	Y		Y
	E6	Y	–	–	–	–
BeiDou	B1I	Y	Y	Y	Y	Y
	B1C	Y	Y	–	–	–
	B2I	Y	Y	Y	Y	Y
	B2a	Y	Y	–	–	–
	B2b	Y	Y	–	–	–
	B3I	Y	–	–	–	–
QZSS	L1 C/A	Y	Y	Y	Y	Y
	L1C	Y	Y	–	–	–
	L2C	Y	Y	Y	Y	Y
	L5	Y	Y	Y	–	Y
	L6	Y	–	–	–	–
NavIC	L5	Y	Y	–	–	–
SBAS	L1	Y	Y	Y	Y	Y
	L5	Y	Y	–	–	–
L-Band		5 channels	5 channels	1 channel	–	1 channel

Table 28: Horizontal position accuracy (RMS)

	PwrPak7 (all models)	FlexPak6	FlexPak6D	ProPak6
Single point L1	1.5 m	1.5 m	1.5 m	1.5 m
Single point L1/L2	1.2 m	1.2 m	1.2 m	1.2 m

	PwrPak7 (all models)	FlexPak6	FlexPak6D	ProPak6
SBAS	60 cm	60 cm	60 cm	60 cm
DGPS	40 cm	40 cm	40 cm	40 cm
TerraStar-L	40 cm	40 cm	–	–
TerraStar-C	–	4 cm	–	4 cm
TerraStar-C PRO	2.0 cm	–	–	–
RTK	1 cm + 1 ppm	1 cm + 1 ppm	1 cm + 1 ppm	1 cm + 1 ppm
RTK ASSIST	Y	Y	–	–
RTK ASSIST PRO	Y	–	–	–

Table 29: Communications ports

		PwrPak7 (all models)	FlexPak6	FlexPak6D	ProPak6
Serial ports	COM1	RS-232/RS-422	RS-232/RS-422	RS-232/RS-422	RS-232/RS-422
	COM2	RS-232/RS-422	RS-232/RS-422	RS-232/RS-422	RS-232/RS-422
	COM3	RS-232	–	–	RS-232/RS-422
	COM7 ¹	–	–	–	RS-232*
	COM8 ¹	–	–	–	RS-232*
	COM9 ¹	–	–	–	RS-232*
	COM10 ¹	–	–	–	RS-232*
USB ports	Device	1	1	1	1
	Host	1	–	–	1
CAN bus ports		1	1	–	2
Ethernet port (10BASE-T/100BASE-TX)		1	1	–	1
Wi-Fi	Access Point	1	–	–	1
	Client	–	–	–	1
Cellular (CDMA or GSM/GPRS/HSDPA)		–	–	–	1
Bluetooth		–	–	–	1
1. Only available when using the ProPak6 Expansion cable. The ProPak6 Expansion cable multiplexes four serial COM ports onto a single USB port.					

Table 30: RF input

		PwrPak7 PwrPak7-E1	PwrPak7D PwrPak7D-E1	FlexPak6	FlexPak6D	ProPak6 dual antenna	ProPak6
GNSS antenna connector	Primary	TNC	SMA	TNC	SMA	TNC	TNC
	Secondary	–	SMA	–	SMA	TNC	–
LNA	Primary	+5 VDC $\pm 5\%$; 200 mA	+5 VDC $\pm 5\%$; 200 mA	+5 VDC $\pm 5\%$; 100 mA	+5 VDC $\pm 5\%$; 100 mA	+5 VDC $\pm 5\%$; 200 mA	+5 VDC $\pm 5\%$; 200 mA
	Secondary	–	+5 VDC $\pm 5\%$; 200 mA	–	+5 VDC $\pm 5\%$; 100 mA	+5 VDC $\pm 5\%$; 100 mA	–
External oscillator connector		–	–	–	–	–	BNC
Cellular antenna connector		–	–	–	–	TNC	TNC

Table 31: I/O strobes

	PwrPak7 (all models)	FlexPak6	FlexPak6D	ProPak6
Event In	3	2	2	4
Event Out	3	1	1	3
PPS	1	1	1	1
ERROR	–	1	–	*
Position Valid (PV)	–	1	1	*
Wheel sensor	1	–	–	–
* The ProPak6 I/O strobes can be configured to change an Event line to this signal.				

Table 32: Features

		PwrPak7-E1 PwrPak7D-E1	PwrPak7 PwrPak7D	FlexPak6	FlexPak6D	ProPak6
Data rates	GNSS measurements	20 Hz	100 Hz	100 Hz	20 Hz	100 Hz
	GNSS position	20 Hz	100 Hz	100 Hz	20 Hz	100 Hz
	INS position/ attitude	200 Hz	*	*	–	*
	INS raw data	125 Hz	*	*	–	*
SPAN compatible		Y	Y	Y	–	Y
Integrated IMU		Y	–	–	–	–

		PwrPak7-E1 PwrPak7D-E1	PwrPak7 PwrPak7D	FlexPak6	FlexPak6D	ProPak6
Compatible UI	NovAtel Application Suite	Y	Y	Y (version 1.1.5.0 or earlier)	Y (version 1.1.5.0 or earlier)	Y (version 1.1.5.0 or earlier)
	Manage Web	Y	Y	–	–	–
Onboard memory		16 GB	16 GB	–	–	4 GB
* INS data rates vary depending on the external IMU connected to the receiver.						

3.6 Differences between SMART7 and SMART6

The following tables compare the specifications and features of the SMART7 with the specifications and features of the SMART6 and SMART6-L.

Table 33: Physical specifications

	SMART7 (all models)	SMART6	SMART6-L
Dimensions	220 x 192 x 66 mm	155 (diameter) x 81 mm	155 (diameter) x 81 mm
Weight	<1.1 kg	<520 g	<570 g

Table 34: Power requirements

	SMART7 (all models)	SMART6	SMART6-L
Input voltage	+7 to +30 VDC	+8 to +36 VDC	+8 to +36 VDC
Power consumption (typical)	4 W	3.5 W	2.9 W

Table 35: Signals tracked

		SMART7 (all models)	SMART6	SMART6-L
GPS	L1 C/A	Y	Y	Y
	L1C	Y	–	–
	L2C	Y	Y	Y
	L2P	Y	Y	Y
	L5	Y	–	–

		SMART7 (all models)	SMART6	SMART6-L
GLONASS	L1 C/A	Y	Y	Y
	L2 C/A	Y	Y	Y
	L2P	Y	–	–
	L3	Y	–	–
Galileo	E1	Y	Y	Y
	E5a	Y	–	–
	E5b	Y	–	–
	E5 AltBOC	Y	–	–
BeiDou	B1I	Y	Y	Y
	B1C	Y	–	–
	B2I	Y	–	–
	B2a	Y	–	–
	B2b	Y	–	–
QZSS	L1 C/A	Y	–	–
	L1C	Y	–	–
	L2C	Y	–	–
	L5	Y	–	–
SBAS	L1	Y	Y	Y
	L5	Y	–	–
L-Band		5 channels	–	Y

Table 36: Horizontal position accuracy

	SMART7 (all models)	SMART6	SMART6-L
Single point L1	1.5 m	1.5 m	1.5 m
Single point L1/L2	1.2 m	1.2 m	1.2 m
SBAS	60 cm	60 cm	60 cm
DGPS	40 cm	40 cm	40 cm
TerraStar-L	40 cm	–	40 cm
TerraStar-C	–	–	4 cm

	SMART7 (all models)	SMART6	SMART6-L
TerraStar-C PRO	2.0 cm	–	–
RTK	1 cm + 1 ppm	1 cm + 1 ppm	1 cm + 1 ppm
RTK ASSIST	–	–	Y
RTK ASSIST PRO	Y	–	–

Table 37: Communications ports

		SMART7 SMART7-S	SMART7-I	SMART7- W	SMART6	SMART6-L
Serial ports	COM1	RS-232	RS-232	RS-232	RS-232	RS-232
	COM2	RS-232	RS-232	RS-232	RS-232	RS-232
	COM3	RS-232	RS-232	RS-232	RS-232	RS-232
CAN bus ports		1	1	1	1	1
Ethernet port 10BASE-T/100BASE-TX		–	1	–	–	–
Wi-Fi	Access Point	–	Y	Y	–	–
	Client	–	Y	Y	–	–
Bluetooth		–	–	–	Y	–

Table 38: I/O strobes

	SMART7 (all models)	SMART6	SMART6-L
Event In	–	1	1
PPS	1	1	1
Emulated Radar	1	1	1

Table 39: Features

		SMART7	SMART7-I SMART7-W	SMART7-S	SMART6	SMART6-L
Data rates	GNSS measurements	20 Hz	20 Hz	20 Hz	20 Hz	50 Hz
	GNSS position	20 Hz	20 Hz	20 Hz	20 Hz	50 Hz
	INS position/ attitude	*	*	200 Hz	–	*
	INS raw data	*	*	125 Hz	–	*
SPAN compatible		Y	Y	Y	–	Y
Integrated IMU		–	–	Y	–	–
Compatible UI	NovAtel Application Suite	Y	Y	Y (version 1.1.5.0 or earlier)	Y (version 1.1.5.0 or earlier)	Y (version 1.1.5.0 or earlier)
	Manage Web	–	Y	–	–	–
* INS data rates vary depending on the external IMU connected to the receiver.						

3.7 Differences between RELAY7 and RELAY

The following tables compare the specifications and features of the RELAY7 with the specifications and features of the RELAY (for the SMART6-L).

Table 40: Physical specifications

	RELAY7	RELAY
Dimensions	186.5 x 199.5 x 76.4 mm	290 x 275 x 80 mm
Weight	<850 g	2.0 kg

Table 41: Power requirements

	RELAY7	RELAY
Input voltage	+7 to +30 VDC	+9 to +36 VDC
Power consumption (typical)	2 W	10.5 W – UHF 400 MHz radio 8.5 W – UHF 900 MHz radio 5.5 W – HSPA radio

Table 42: Supported radios

	RELAY7	RELAY
400 MHz	Y	Y
900 MHz	Y	Y
HSPA	–	Y
Wi-Fi	–	Y
The RELAY7 will have either a 400 MHz or 900 MHz radio.		
The RELAY will have Wi-Fi and either a 400 MHz, 900 MHz or HSPA radio.		

Chapter 4 Receiver technical specifications

Refer to the following documents for the technical specifications of OEM7 receivers.

- [OEM7 Installation and Operation User Manual](#) – OEM719, OEM729, OEM7600, OEM7700 and OEM7720
- [PwrPak7 Installation and Operation User Manual](#) – PwrPak7, PwrPak7D, PwrPak7-E1, PwrPak7D-E1, PwrPak7-E2 and PwrPak7D-E2
- [CPT7 Installation and Operation User Manual](#) – CPT7 and CPT7700
- [SMART7 Installation and Operation User Manual](#) – SMART7 and SMART7-S
- [SMART2 Installation and Operation User Manual](#) – SMART2

